Public Provision and Private Competition in Health Insurance*

Jinglin Wang†

November 8, 2026

FREQUENTLY UPDATED

CLICK HERE FOR THE LATEST VERSION

Abstract

Many social programs provide public benefits that can reduce demand for private options. Expanding eligibility for these programs can create a trade-off between broadening access to public benefits and preserving the demand that sustains private competition and consumer choice. I study this trade-off in the U.S. health insurance market. Using administrative enrollment and claims data from Colorado, I develop and estimate a structural model of insurance demand, pricing, and insurer participation to evaluate an expansion of Medicaid eligibility from 138 to 200 percent of the federal poverty level. I find the expansion removes relatively low-cost enrollees from the private market. Adverse selection is limited, however, because premium subsidies largely offset higher gross premiums. The policy does reduce private demand, lowering insurer participation and leaving remaining consumers with fewer coverage options. Welfare losses among these consumers are concentrated in smaller markets, where expansion leads to a greater decline in the probability of having multiple insurers. These findings highlight the role of insurer participation in shaping the welfare effects of public coverage expansions and inform recent proposals to extend Medicaid eligibility.

*I am grateful to Michael Dickstein, Daniel Waldinger, Chris Conlon, and Elena Manresa for invaluable guidance and support. I also thank Giulia Brancaccio, Jeff Gortmaker, Felix Montag, Thi Mai Anh Nguyen, Paul Scott, Audrey Tiew, and seminar participants at the NYU Applied Microeconomics and Industrial Organization workshops for insightful comments and suggestions. I gratefully acknowledge financial support from the NYU Center for Applied and Theoretical Econometrics and the C.V. Starr Center for Applied Economics. All errors are my own.

†Department of Economics, New York University. Email: jinglin.wang@nyu.edu.

1 Introduction

Many social programs provide public benefits that can substitute for private options. Extending eligibility broadens access to public support but also reduces demand for private options. This demand contraction may induce firm exit, weakening competition and choice for those who continue to rely on private markets. Selection markets such as health insurance face an additional concern: changes in the composition of private demand can worsen adverse selection, raising costs and prices and potentially unraveling the market.

In the United States, lower-income adults can get public health insurance coverage through Medicaid, with an income eligibility cutoff of 138 percent of the federal poverty level (FPL) in most states. The main alternative to Medicaid is the private individual market, where consumers purchase plans directly from insurers. Whether to extend public coverage remains an active policy question: Oregon introduced coverage up to 200 percent FPL in 2024, and Colorado is evaluating a similar extension (Oregon Health Authority, 2024; Colorado General Assembly, 2026). Policy discussions have identified concerns about private-market stability and insurer participation but do not quantify the effects of broader eligibility on private markets and consumer welfare (Dorn and Tolbert, 2014; Corlette et al., 2023).

This paper evaluates how extending Medicaid eligibility affects the private individual market and the welfare of consumers who remain ineligible for public coverage. Because Medicaid eligibility generally precludes access to private premium subsidies, extending eligibility can shift consumers out of the private market, leading to concerns about adverse selection and private insurer participation. To quantify insurers' responses, I develop and estimate a model in which insurers, differentiated by provider networks, endogenously choose pricing and market participation. This framework links insurers' participation decisions to consumers' access to both coverage and health care providers. In counterfactual simulations, broader eligibility reduces welfare primarily through insurer exit, but premium subsidies limit further adverse selection.

I estimate the model using administrative enrollment and claims data from Colorado and evaluate an extension of Medicaid eligibility from 138 to 200 percent FPL. The data link consumers' insurance choices to medical utilization and spending, informing the estimation of demand and insurer costs. Colorado provides a useful setting for studying participation responses because its

Medicaid program pays for most physical health services through a publicly administered fee-for-service system (Group, 2019). This limits individual-market insurers' ability to offset lost private enrollment by serving the same consumers through Medicaid managed-care plans. Colorado's counties also vary substantially in population, enrollee composition, and insurer participation, allowing me to examine how the consequences of broader public eligibility differ across local markets.

I begin by documenting four descriptive patterns that motivate the analysis. First, current individual-market enrollees with assigned incomes of 138–199 percent of FPL—the expansion group—have lower medical spending than other enrollees. Holding other enrollment fixed, their departure would raise average spending in the remaining private risk pool, creating the potential for higher premiums and further adverse selection. Second, comparisons across counties reveal a strong positive correlation between medical spending and gross premiums, but little correlation between spending and net premiums. This pattern is consistent with premium subsidies dampening the feedback between higher costs and further adverse selection in the individual market. Third, the expansion group accounts for a considerable share of current private enrollment, suggesting that its departure could weaken insurers' participation incentives even when subsidies limit adverse selection. Finally, smaller markets offer fewer insurance options, and insurer exit can also reduce access to hospital networks. Removing each enrollee's current insurer shows that a sizable share of those with an alternative insurer would lose in-network access to the nearest hospital in their original network. These patterns suggest that changes in insurer participation could affect both coverage availability and hospital access.

To quantify the welfare and government spending consequences of extending Medicaid eligibility, I build and estimate a model of insurance demand and supply with insurer market participation as a central margin of adjustment. On the demand side, consumers choose among insurance plans based on premiums net of subsidies, coverage generosity, and hospital networks. I model hospital demand to measure consumers' valuations of insurer networks based on their expected utilization of hospital care (Ho, 2006; Ho and Lee, 2017). The network measure allows for both vertical differentiation in network quality and horizontal differentiation in how well networks match consumers' locations and anticipated medical needs. Together with insurance preferences that vary by spending risk, it also captures how consumers with different health risks sort

across insurers.

On the supply side, insurers choose which markets to serve and compete in premiums subject to a regulatory constraint on markups. Insurers set premiums to maximize profits, accounting for how their prices and networks relative to rivals affect both enrollment and its risk composition. I model annual participation as a static game of incomplete information (Seim, 2006): insurers participate when expected variable profits exceed their fixed costs, accounting for uncertainty about rivals' participation. The static approximation is motivated by insurers' ability to maintain local provider networks through their other insurance business, which reduces the effect of current individual-market participation on future participation costs. In this static formulation, fixed costs largely reflect the cost of maintaining provider networks, and I allow fixed costs to differ by prior participation to account for the cost advantages of existing local infrastructure. This interpretation also motivates the incomplete-information assumption, as competitors are unlikely to observe the cost of maintaining local networks.

The counterfactual results show that market features introduced by the Affordable Care Act (ACA)—premium subsidies and medical loss ratio (MLR) regulation— mitigate some concerns about broader Medicaid eligibility for remaining private-market consumers. Although the departure of lower-cost enrollees raises gross premiums, subsidies rise with premiums and leave average net premiums largely unchanged, limiting further unraveling due to adverse selection. MLR regulation also constrains markups as insurer participation declines and markets become more concentrated. As a result, when insurer participation probabilities are held at baseline, consumer surplus changes little.

However, these protections do not eliminate the welfare costs of insurer exit. Broader Medicaid eligibility reduces private enrollment and expected profits, weakening insurers' incentives to participate. Their withdrawal reduces coverage choice and access to valued provider networks even when net premiums remain stable. These effects vary widely across local markets. In the smallest markets, the probability of having multiple insurers falls by 15.6 percent (8.6 percentage points), while larger markets generally retain multiple insurers. Annual consumer surplus falls by $115.78 per remaining person in the smallest third of counties and $28.14 in the middle third, while remaining essentially unchanged in the largest third.

This analysis illustrates a broader distinction between preserving the affordability of private

options and preserving their availability. When public benefits draw consumers away from private providers, policies that protect remaining consumers from higher prices may not prevent exit as providers face shrinking demand. The resulting welfare consequences depend on how readily consumers can substitute for the options lost. These lessons extend to other settings where public benefits substitute for differentiated private services, highlighting the availability and value of remaining private options as key considerations in evaluating broader public eligibility.

Relation to Literature This paper directly contributes to the literature on how public insurance expansions affect private insurance markets. Cutler and Gruber (1996) and Gruber and Simon (2008) study whether expansions in Medicaid eligibility crowd out private coverage demand. Clemens (2015), Peng (2017), and Sen and DeLeire (2018) examine how Medicaid expansion changes the composition and premiums of the private individual market. I extend this literature by examining how expansion affects coverage options and welfare for consumers who remain ineligible for public insurance. Closest to this paper, Kanimian (2025) studies whether Medicaid eligibility thresholds should reflect geographic differences in the cost of living, allowing Marketplace insurers to adjust premiums and Medicaid plans to adjust quality while holding insurer participation fixed. I instead study a uniform eligibility expansion with endogenous private insurer entry and exit. This allows me to quantify an additional consequence of expanding public coverage: the loss of private coverage options for consumers who do not gain access to the public program.

This paper also connects to research on market segmentation in health insurance. Dickstein, Ho and Mark (2024) study pooling the small-group and individual markets, allowing insurers to reprice while holding plan offerings fixed. Fleitas et al. (2025) examine policies that move households from the small-group market into the Marketplace. These papers highlight how the allocation of consumers across separately regulated insurance pools shapes selection and welfare. I examine this question at the Medicaid–individual market boundary, allowing changes in the allocation of consumers to alter which insurers participate in the remaining private market. My analysis highlights how an eligibility reform could change not only the terms of available coverage but also whether that coverage remains available.

Finally, this paper relates to the literature on how selection, market power, and regulation

shape insurance markets (Starc, 2014; Ericson and Starc, 2015; Mahoney and Weyl, 2017; Jaffe and Shepard, 2020; Shepard, 2022; Tebaldi, 2025). Within this literature, this paper joins a small group of recent papers that focus on insurer participation as a key channel affecting market outcomes. Geddes (2025) studies how geographic pricing rules affect entry and coverage availability, while Zahn (2025) examines how Medicare Advantage insurers use entry and product offerings to select healthier beneficiaries from a population already eligible for public coverage through traditional Medicare. I study how extending public eligibility contracts private demand and affects insurer participation and welfare for consumers who remain ineligible for public coverage. I also complement these studies by explicitly modeling horizontal differentiation in hospital networks and its implications for risk sorting and the welfare costs of insurer exit.

The rest of this paper is organized as follows. Section 2 describes the institutional background. Section 3 describes the data used for this paper and shows descriptive facts that illustrate the market characteristics driving the main welfare conclusions. Section 4 presents the model and describes the estimation procedure and results. Section 5 simulates and interprets the counterfactual outcomes. Section 6 concludes.

2 Institutional Background

Medicaid Medicaid is the largest source of funding for medical and health-related services for people with low income in the United States, including children, pregnant individuals, older adults, and people with disabilities. In December 2025, Medicaid covered approximately 68.5 million individuals nationwide (Centers for Medicare & Medicaid Services, 2026).

Each state administers its Medicaid program subject to federal requirements. Colorado's Medicaid program, Health First Colorado, covers over one million adults and children each year. Eligibility is primarily income-based. In 2014, Colorado implemented its Medicaid expansion program to extend coverage to nearly all non-elderly adults with household incomes up to an effective threshold of 138% of the Federal Poverty Level. Medicaid enrollees in Colorado receive comprehensive benefits including physical, behavioral, and dental health services alongside long-term supports. Enrollees pay zero premiums and pay little, if anything, out-of-pocket, for most services. Eligibility for Medicaid generally precludes eligibility for government subsidies towards Marketplace coverage. Operationally, most health services are reimbursed fee-for-service at state-

determined rates, with regional organizations coordinating care and maintaining provider networks.[1]

The Individual Market The U.S. individual health insurance market provides coverage for people who do not obtain health insurance through an employer or a public program such as Medicare or Medicaid. As of the first quarter of 2025, an estimated 25.2 million people were enrolled in the individual market (KFF, 2025).

Colorado operates its own ACA marketplace, Connect for Health Colorado, through which consumers compare and purchase private plans. Consumers can also purchase individual coverage off the Marketplace, but federal premium tax credits and cost-sharing assistance are available only through the Marketplace. My analysis covers both channels during 2017–2019.

Plans offered in the individual market are standardized in several ways. All plans are mandated to cover essential health benefits such as preventive services and mental health benefits. The cost-sharing features of the plans, such as deductibles and coinsurance, are summarized by different metal levels, so that bronze plans cover roughly 60% of medical costs, silver plans cover 70%, gold plans cover 80%, and platinum plans cover 90%. During my sample period, insurers participating in Colorado's exchange were required to offer at least one silver plan and one gold plan in each service area (U.S. Department of Health and Human Services, n.d.). No insurer in my analysis offers platinum plans.

Individual market insurers are also subject to federal and state pricing regulations that constrain how premiums are set. Under community rating rules, premiums cannot vary based on individual health status or prior claims experience, and can only vary within limits based on factors such as age and geography. Within a rating area, premiums for the same plan may vary by age according to a statutory schedule and by self-reported tobacco use (Colorado Division of Insurance, 2018; Centers for Medicare & Medicaid Services, 2013). Insurers must also satisfy medical loss ratio (MLR) requirements, which mandate that a minimum share of premium revenue be spent on medical claims and qualifying activities that improve health care quality (Centers for Medicare & Medicaid Services, n.d.*c*). This restriction provides an additional constraint on insurers' pricing

[1]This applies to physical health services and long-term care services. For behavioral health services, the regional organizations receive a fixed, monthly per-member budget and negotiate independently with local mental health providers to determine the reimbursement fee schedule. See Group (2019) for the 2018–2019 RAE payment structure and its managed-care exceptions.

responses as enrollee costs change.

Insurer entry in the individual market is typically determined at the county level and is made on a plan-year basis. Insurers make annual decisions to enter, exit, or continue participation in each market several months prior to the start of the coverage year. Therefore, insurance availability can differ between counties and between years in the same county.

Households can purchase insurance coverage in the individual market either through a marketplace created under the ACA or through an off-marketplace channel, but households eligible for federal financial assistance must enroll through the Marketplace to receive those benefits. The primary form of assistance is the premium tax credit (PTC), which reduces monthly premium payments for eligible households. During my sample period, eligible households with incomes up to 400 percent FPL could receive premium tax credits.[2] The credit covers the positive difference between the premium of the second-cheapest Silver plan available to the household and an income-based required contribution, capped at the premium of the selected plan. Because subsidies adjust with this benchmark, changes in gross premiums need not translate into comparable changes in consumers' net payments.

In addition to premium subsidies, lower-income households enrolling in Silver plans may qualify for cost-sharing reductions (CSRs), which lower out-of-pocket expenses. Unlike premium tax credits, which reduce monthly premiums, CSRs increase the actuarial value of eligible Silver plans by requiring insurers to offer plan variants with lower out-of-pocket costs in a way that satisfies a federally specified actuarial value. During my study period in Colorado, Marketplace enrollees with incomes between 138% and 150% of the federal poverty level qualified for Silver plans with a 94% actuarial value, while those with incomes between 150% and 200% and between 200% and 250% of the federal poverty level qualified for plans with actuarial values of 87% and 73%, respectively (U.S. Department of Health and Human Services, 2025).

[2] At 138% of the federal poverty level, the required contribution toward the benchmark premium was approximately 3.36%, 3.32%, and 3.42% of income in 2017, 2018, and 2019, respectively. At 200% FPL, the corresponding percentages were 6.43%, 6.34%, and 6.54%; at 300–400% FPL, they were 9.69%, 9.56%, and 9.86%. The 138% FPL values use linear interpolation within the published income bands. See Internal Revenue Service (2016, 2017, 2018).

3 Data and Descriptive Patterns

3.1 Data

The main dataset in this paper is the Colorado All-Payer Claims Database (CO APCD), an administrative dataset that captures nearly all insurance enrollment and claims for Colorado residents (Center for Improving Value in Health Care, 2026). For each member, the dataset includes information on their insurance plan choice, the plan's coverage start and end dates, whether the purchase was made through the ACA exchange, and the member's age, gender, race, and residence location (5-digit zip code). For each claim record, the data include information on the procedure(s) filed, the amount paid by the patient and the insurer, and the facility or provider associated with the reported service.

The insured analysis population is constructed as repeated March cross-sections of the Colorado individual market from the CO APCD for 2017–2019.[3] I exclude insurers operating only off-exchange, which account for approximately 3% of individual-market enrollment in Colorado over 2017–2019.[4] The analysis sample is restricted to 543,431 member-years for adults aged 18–64 who were enrolled on March 1 in non-grandfathered individual-market coverage, including both on- and off-exchange enrollees of the included insurers. I further restrict the sample to 535,575 member-years with a unique nonmissing zip-age-sex combination. Finally, I exclude 163 member-year records with more than one insurer-metal record, leaving a final sample of 535,412 member-years.

I summarize health-related differences across enrollees using a spending-risk score that predicts log spending on medical and pharmacy services per covered month from age, sex, and diagnoses recorded during the same year.[5] I classify enrollees as low risk if their score falls below the median among eligible person-years pooled over 2017–2019, and as high risk otherwise, using a common cutoff across years. Appendix C.1 describes the prediction procedure and its validation.

[3] The open enrollment period for the Colorado exchange ends in January of each year, and coverage typically begins in February at the latest. I use March enrollment data because some insurers appear to have enrollment data reporting lags in February. I treat March enrollment as representative of year-round enrollment. While enrollment patterns could differ by month, incorporating enrollment timing decisions requires substantially more data and modeling and is outside the scope of this paper (see Diamond et al., 2026).

[4] Author's calculation from the 2017–2019 CMS medical loss ratio public use files (Centers for Medicare & Medicaid Services, n.d.*b*).

[5] Using same-year information allows me to retain enrollees without observable prior-year claims histories. High turnover in the individual market limits the sample available when risk is measured using lagged diagnoses.

The CO APCD does not report income, which determines eligibility for public coverage and premium subsidies. Although exchange applications collect income information to determine financial assistance, they do not provide a common income measure covering off-exchange enrollees and uninsured adults. To retain detailed enrollment and health information in the claims while preserving uncertainty about individual income, I combine administrative records with external income distributions, assign each insured enrollee probabilities over six income categories using coverage-specific income priors, and condition on county, age, sex, and predicted spending-risk group.[6] Among consumers assigned incomes of 0–137 percent of FPL, I treat observed individual-market enrollees as Medicaid-ineligible and uninsured adults as Medicaid-eligible individuals who do not take up coverage.[7]

The assignment combines coverage-specific statewide income distributions with assumptions about geographic and health-related variation. Statewide income shares are anchored to Colorado exchange enrollment reports (Connect for Health Colorado, 2019) for on-exchange enrollees and the Current Population Survey Annual Social and Economic Supplement (CPS ASEC) (U.S. Census Bureau, n.d.*c*) for off-exchange enrollees. The assignment relies on two main assumptions. First, I assume that relative income differences across county-age groups measured in the American Community Survey (ACS; U.S. Census Bureau, n.d.*b*; n.d.*a*) apply within each coverage group. This allows local income composition to vary while preserving differences in statewide income distributions between on- and off-exchange enrollees. Second, I use the income–spending-risk relationship among respondents to the Medical Expenditure Panel Survey (MEPS) in the Western region (Agency for Healthcare Research and Quality, n.d.), separately for insured and uninsured

[6]The categories are 0–137, 138–199, 200–249, 250–299, 300–399, and 400 percent or more of FPL. They isolate the population newly eligible under the expansion and retain selected thresholds relevant to Medicaid eligibility and marketplace assistance during 2017–2019. In particular, 200, 250, and 300 percent mark boundaries in the premium-contribution schedule, and 400 percent marks the historical upper income limit for premium tax credits. The categories also broadly follow marketplace enrollment reporting bands; differences in source-data classifications are reconciled as described in Appendix A.

[7]Although adults below 138 percent of FPL are generally eligible for Medicaid in Colorado, immigration-related restrictions exclude some, including certain lawfully present immigrants who may instead qualify for Marketplace subsidies (Colorado Health Institute, 2021, p. 1). Medicaid eligibility determines whether these consumers belong in the private-market welfare and subsidy evaluations. If some private enrollees are Medicaid-eligible, they should be excluded, and the model misspecifies their Medicaid outside option. Choosing unsubsidized private coverage would, however, require coverage advantages or Medicaid enrollment barriers sufficient to justify full premiums, a demanding condition given the cost of unsubsidized coverage. Conversely, Medicaid-ineligible uninsured consumers should be included in the private welfare/subsidy evaluations, although expansion would leave their eligibility unchanged. Their exclusion would not affect estimated policy effects if they remain uninsured, but could omit welfare gains and subsidy spending if improved private options induce enrollment. I plan to assess sensitivity to both eligibility assumptions in future revisions.

adults and conditional on broad age group and sex. Because MEPS jointly observes income, demographics, and diagnoses, it supplies the income–risk relationship.

I augment the CO APCD with a sample of uninsured individuals by constructing a synthetic uninsured population based on the insured population. To include uninsured adults, I apply ACS uninsured-to-direct-purchase coverage ratios to the corresponding county-age cells in the CO APCD sample, and assign each observation a joint income-risk distribution based on the uninsured MEPS relationship. Appendix A provides the construction details.

Table 1 reports summary statistics for the potential private-market population, comprising observed individual-market enrollees at all income levels and uninsured consumers with assigned incomes of at least 138 percent of FPL. Within each year, counties are divided into thirds by the size of this population, before network-support exclusions. Population is highly concentrated: the largest third accounts for 93.3 percent of potential consumers, compared with 1.4 percent in the smallest third. Smaller markets have older populations but lower shares of high-risk consumers. They also have larger shares of consumers with assigned incomes of 138–199 percent of FPL: 30.5 percent in the smallest third, compared with 22.2 percent in the largest third. Private insurance enrollment rates are lower in both smaller groups, and insurer participation differs substantially across groups, averaging 1.50 active insurers in the smallest third and 3.17 in the largest.

Table 1: Sample characteristics by market size, 2017–2019

	All markets	Smallest third	Middle third	Largest third
Number of county-years	183	60	60	63
Median market size	622.9	147.4	601.7	5,246.4
Share of potential consumers (%)	100.0	1.4	5.4	93.3
Mean age	44.1	47.0	45.9	43.9
High-risk share (%)	33.8	31.1	31.4	34.0
Assigned income 138–199% of FPL (%)	22.5	30.5	27.2	22.2
Private insurance enrollment rate (%)	55.0	52.6	51.6	55.2
Mean number of active insurers	2.16	1.50	1.77	3.17

Notes: The sample comprises counties observed in 2017–2019. Market size includes observed individual-market enrollees at all income levels and uninsured consumers with assigned incomes of at least 138 percent of FPL. Counties are ranked by this population separately within each year. Population characteristics use population weights; median market size and mean insurer count give each county-year equal weight. High risk denotes a spending-risk score at or above the pooled median among insured person-years. An insurer is active with at least five recorded March on-exchange enrollees.

Table 2 summarizes assigned income and spending risk in this potential private-market population. Consumers with assigned incomes between 138 and 199 percent of FPL, who would become eligible for Medicaid under the expansion, account for 22.5 percent of this population. Within this group, 24.8 percent are classified as high risk, compared with 33.8 percent overall. The high-risk share is approximately 32–33 percent among consumers with assigned incomes of 200–399 percent of FPL and rises to 38.7 percent at or above 400 percent. The 0–137 percent group includes only observed private enrollees, of whom 51.9 percent are high risk.

Table 2: Assigned income and spending risk, 2017–2019

Income (% of FPL)	Share of potential population (%)	High-risk share within income group (%)
0–137	8.7	51.9
138–199	22.5	24.8
200–249	16.6	32.6
250–299	12.7	32.0
300–399	17.9	32.8
400 or more	21.5	38.7
All incomes	100.0	33.8

Notes: The sample pools counties over 2017–2019. The potential population includes observed individual-market enrollees at all income levels and uninsured consumers with assigned incomes of at least 138 percent of FPL. The 0–137 percent row includes private enrollees only. High risk denotes a spending-risk score at or above the pooled median among eligible insured person-years.

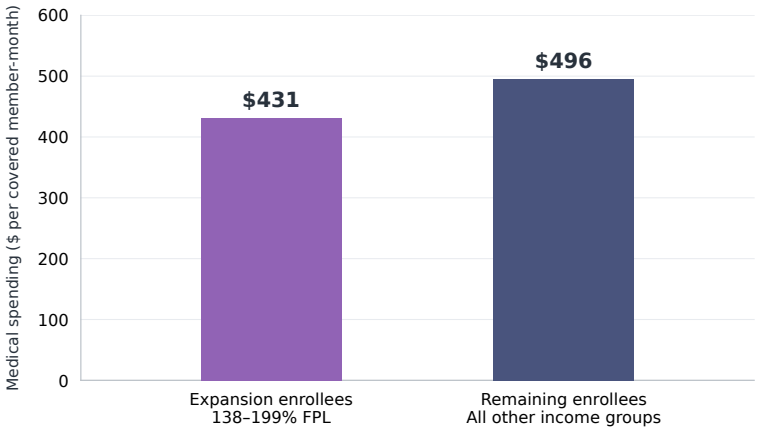
3.2 Descriptive Patterns

This section presents descriptive patterns across Colorado's individual insurance markets relevant to the key welfare conclusions, using the income assignment described previously. These patterns suggest that evaluating the consequences of the eligibility expansion requires accounting for the market's subsidy-setting mechanism and insurers' participation incentives.

I first compare medical spending among current enrollees who would become eligible for Medicaid under the expansion with spending among all other current enrollees. Figure 1 shows that spending averages $431 per covered member-month among enrollees with assigned incomes between 138 and 199 percent of FPL, compared with $496 among remaining enrollees—a difference of approximately 13 percent. Holding other enrollment fixed, the departure of this lower-spending group would raise average observed spending in the remaining private risk pool, creat-

ing potential upward pressure on gross premiums.

Figure 1: Medical spending by expansion eligibility



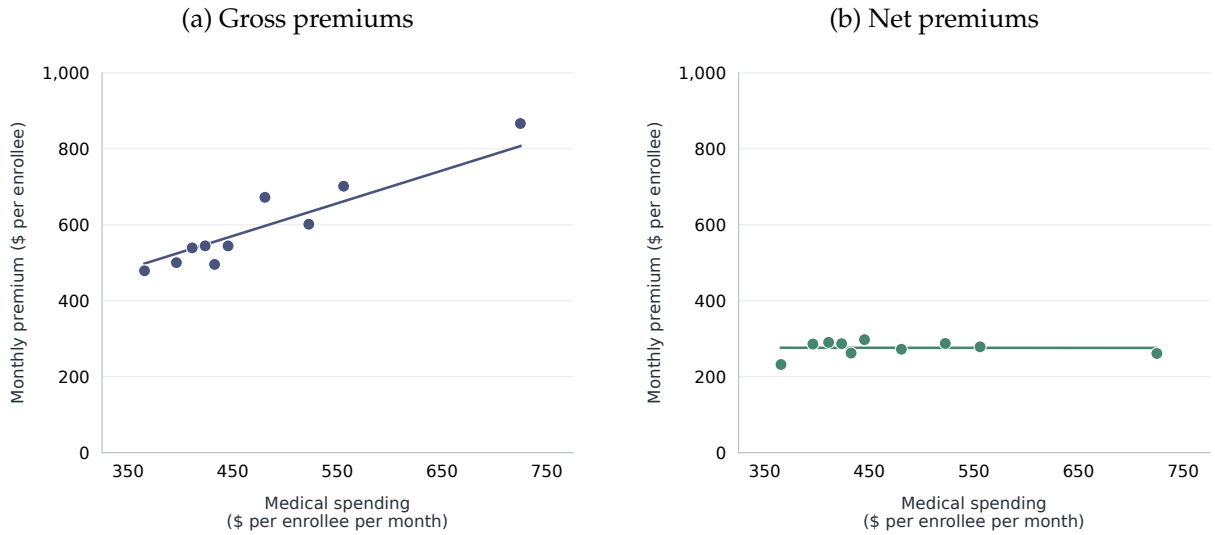
Notes: The figure pools 2017–2019 individual market enrollees in Colorado. Expansion enrollees have assigned incomes of 138–199 percent of FPL; remaining enrollees include all other income groups. Each mean divides total medical and pharmacy claims including patient liability by total payer-member months.

This composition change could lead to further adverse selection if higher premiums induce relatively healthy consumers to leave private coverage, but the strength of this feedback depends on the premiums consumers pay after subsidies. Figure 2 shows that higher costs may raise insurers' gross premiums without producing a comparable increase in the prices consumers face. Panels (a) and (b) each relate county-year average medical spending across current enrollees to average gross and net premiums. Gross premiums are strongly positively correlated with spending (weighted correlation 0.632), whereas the correlation between net premiums and spending is close to zero (-0.003). Consistent with subsidies dampening premium-driven adverse selection, higher gross premiums in higher-spending markets tend to be accompanied by larger subsidies, leaving average net payments largely unrelated to spending.

Even if subsidies dampen premium-driven selection, the expansion could still affect insurers' profits by reducing their enrollment. Figure 3 shows that newly eligible consumers account for 17.8 percent of current private enrollment overall. Grouping counties into thirds within each year by their potential individual-market population, this share is 24.7 percent in the smallest markets, 21.9 percent in middle-sized markets, and 17.5 percent in the largest markets. These shares indicate substantial exposure to enrollment loss, particularly in smaller markets.

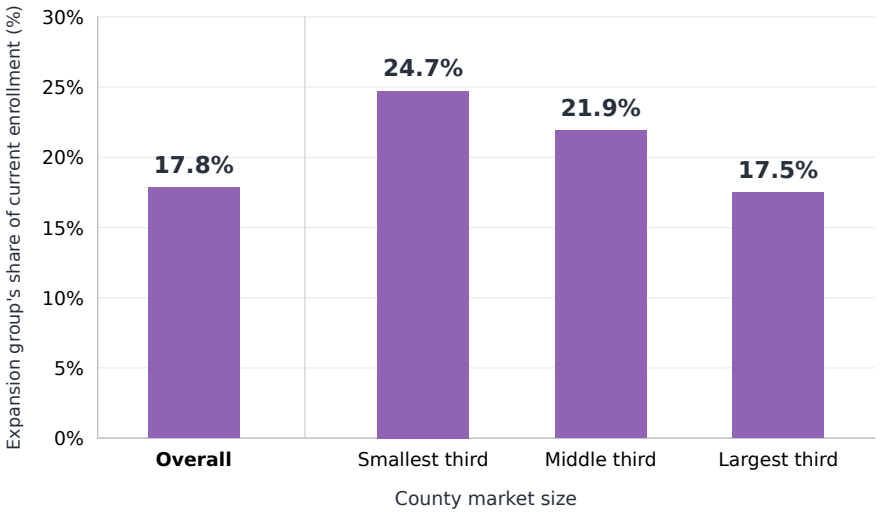
To understand how changes in insurer participation would affect consumers, Figure 4 exam-

Figure 2: Medical spending and gross vs net premiums



Notes: Sample includes current individual market enrollees in all income groups. Spending is each enrollee's annual medical and pharmacy claims, including patient liability, divided by their covered member months. Gross premiums are age-rated premiums. Net premiums subtract subsidies computed under the year's benchmark and contribution rules. Dots report enrollment-weighted means within ten spending bins containing approximately equal current enrollment. Lines are fitted to the unbinned county-year means weighted by enrollment. Both panels use identical bins and axes. Dollar amounts are nominal.

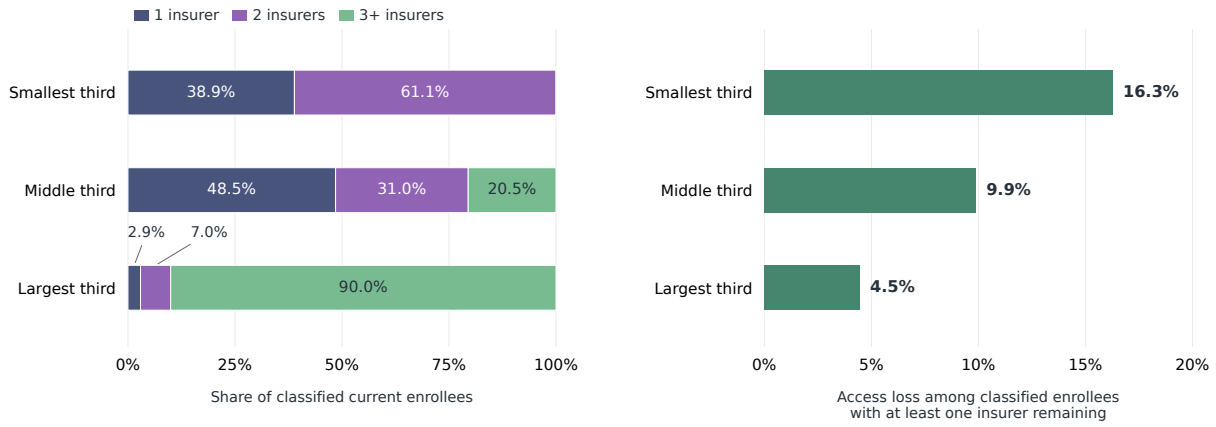
Figure 3: Exposure of private enrollment to Medicaid expansion



Notes: The figure pools 2017–2019 individual market enrollees in Colorado. Expansion-group enrollment is the mass assigned incomes of 138–199 percent of FPL. Each bar divides expansion-group enrollment by all current private enrollment pooled across counties and years. Counties are grouped into thirds separately within each year by the potential private-market population.

ines current insurance availability and the consequences of removing enrollees' current insurer for access to nearby in-network hospitals. Availability of insurance options is highly uneven across markets. Panel (a) of Figure 4 shows that all enrollees in the smallest markets face only one or two insurers, while 90 percent in the largest markets have at least three. A single exit in the smallest markets would therefore leave either a monopoly or no insurer. Because insurers offer different networks, insurer participation also matters for health care access, even when multiple insurers remain in the market. Panel (b) examines changes in in-network hospital access when at least one other insurer remains available. If each enrollee's current insurer were removed separately and other insurers' participation and networks held fixed, 16.3 percent of enrollees in the smallest markets who retain at least one alternative insurer would lose in-network access to their nearest currently in-network hospital. The corresponding shares are 9.9 percent in middle-sized markets and 4.5 percent in the largest markets. Thus, changes in insurer participation could affect both coverage availability and network access.

Figure 4: Insurer participation and hospital access by market size



(a) Available insurers

(b) Hospital access loss conditional on an alternative insurer remaining

Notes: Both panels pool 2017–2019 individual-market enrollees. Counties are grouped into thirds separately within each year by the potential private-market population. An insurer is active if it has at least five recorded March on-exchange enrollees in a county-year. Panel (a) reports observed insurer availability among current enrollees. Panel (b) removes each enrollee's current insurer separately, holding other insurers' participation and networks fixed. Access loss occurs when no remaining insurer covers the nearest hospital in the enrollee's current network. Enrollees in monopoly markets are excluded from Panel (b).

4 Model and Estimation

In this section, I develop a model of insurer participation, premium setting, and consumer demand to evaluate the welfare consequences of Medicaid expansion. The model proceeds in three stages. First, insurers observe private fixed-cost shocks and decide whether to enter a county, accounting for uncertainty about rivals' participation and subsequent marginal-cost realizations. Second, participating insurers observe each other's marginal cost shocks and compete in premiums à la Bertrand–Nash, subject to the regulated markup constraint.[8] Third, consumers choose among the available insurance plans based on plan premiums, coverage, and their valuation of the plan's network based on their expected utilization.

4.1 Demand

I model consumer utility from insurance as a function of the plan's premium net of subsidies, actuarial value, and provider network to allow for both vertical and horizontal competition between insurers. The network component captures the consumer's expected use of hospital care and access to nearby, in-network hospitals offering services relevant to their anticipated medical needs (Ho, 2006; Ho and Lee, 2017; Shepard, 2022).[9] I measure the network component as the utility from hospital care a consumer expects across the diagnoses they may experience. I first describe hospital choice conditional on a diagnosis, then derive the expected network value that enters insurance demand. Conditional on diagnosis d, individual i, insured by insurer f, receives utility from hospital h that depends on a hospital-year quality term η, distance, the hospital's in-network status, and the number of services relevant to that diagnosis available at the hospital:

$$u^H_{ifdht} = \eta_{ht} + \lambda^D_{g(i)} \,\text{Distance}_{ih} + \lambda^N_d \,\text{InNetwork}_{fht} + \sum_s \lambda^S_s \mathbf{1}\{d = d(s)\} \,\text{Service}_{hs} + \nu_{ifdht}.$$

Distance preferences vary across hospital referral regions g to capture geographic differences in the willingness to travel for care, and network preferences vary by diagnosis to capture differences

[8] The full-information pricing equilibrium approximates an outcome in which insurers have incorporated competitors' premium responses into their own pricing decisions. This interpretation does not require competitors' costs to be directly observed, but it abstracts from remaining strategic uncertainty about competitors' prices, which is plausible in this setting where the regulatory review process publicly discloses proposed premiums.

[9] I focus on hospital networks as the measure of network value because claims data for individual physicians and practices are too sparse to construct a measure of physician networks. Hospital network value is likely highly correlated with the strength of the local physician network.

in the value of in-network access across diagnoses; ν_{ifdht} is an independent Type-I extreme-value shock.

The estimated hospital utility allows me to compute expected utility from the best hospital choice conditional on diagnosis, $\mathbb{E}_\nu\left[\max_h \widehat{u}^H_{ifdht}\right]$. Network value then aggregates expected hospital utility across diagnoses:

$$\text{NetworkValue}_{ift} = \sum_d \gamma_i^d \, \mathbb{E}_\nu\left[\max_h \widehat{u}^H_{ifdht}\right],$$

where $\widehat{u}^H$ is estimated hospital utility, and γ_i^d is the probability of the diagnosis in i's age-sex group. Appendix B.1 provides details on the model and estimation of the network measure.

This measure of expected network value enters consumers' insurance utility alongside net premiums and actuarial values. I allow preferences for these plan attributes to vary with predicted spending risk, summarized by a score from a predictive model of medical expenditure based on demographics and diagnosis data. To keep the model tractable, I divide consumers into two risk groups indexed by r using the median predicted score, and estimate separate preference parameters for each group. Appendix C.1 describes the score construction and assesses its predictive performance.

The utility individual i in market m and year t gets from plan j sold by firm $f(j)$ is given by

$$\begin{aligned} u^{(b)}_{ijmt} = {} & \alpha_r \, \text{NetPremium}^{(b)}_{i,j,m,t} + \beta^{AV}_r \, \text{ActuarialValue}^{(b)}_{j,m,t} + \beta^N_r \, \text{NetworkValue}_{i,f(j),m,t} \\ & + \delta_{r(i),f(j),m,t} + \epsilon_{ijmt} \end{aligned} \tag{1}$$

where b is a (latent) income bin; $\text{NetPremium}^{(b)}_{i,j,m,t}$ is the net premium a consumer pays for plan j after accounting for income-specific subsidy, measured in hundreds of dollars per month.[10] $\text{ActuarialValue}^{(b)}_{j,m,t}$ is the share of medical spending expected to be covered by plan j. This term is also income specific because those with income below 250% FPL qualify for cost-sharing reductions when they enroll in any Silver-tier plan.[11] $\text{NetworkValue}_{i,f(j),m,t}$ is the expected network

[10] $\text{NetPremium}^{(b)}_{i,j,m,t} = \max\{0, \text{GrossPremium}_{i,j,m,t} - \text{Subsidy}_{i,m,t}(b)\}$, where $\text{Subsidy}_{i,m,t}(b)$ is determined according to a statutory income schedule that decreases with income.

[11] During my sample period, cost-sharing reductions raised the actuarial value of Silver plans from 70% to 94% for eligible enrollees with incomes up to 150% FPL, 87% for those above 150% and up to 200% FPL, and 73% for those above 200% and up to 250% FPL.

value defined above. The fixed effects $\delta_{r(i),f(j),m,t}$ vary by risk group, insurer, county, and year and capture other plan characteristics unobservable to the econometrician. Finally, ϵ_{ijmt} is the standard logit utility shock following a Type 1 Extreme Value distribution.

Drivers of selection In this model, selection is driven by several forces. First, differences in premium sensitivity across spending-risk groups also allow selection along the income margin. Although the premium coefficient is common across incomes within each risk group, the assigned joint distribution of income and risk gives income groups different mixtures of preference types. As Table 2 shows, consumers newly eligible for Medicaid have a lower high-risk share than consumers remaining in the private market. If higher-risk consumers are less sensitive to net premiums, this difference in risk composition implies a less negative average net-premium coefficient among those remaining. Additionally, income-based subsidies affect how changes in gross premiums pass through to the net premiums facing different income groups. Hence, marginal consumers gained following a premium reduction may differ from existing enrollees in both expected costs and income. Second, preferences for actuarial value may differ across spending-risk groups, allowing plans with different coverage generosity to attract different risk pools, while income-based cost-sharing reductions will additionally shape sorting by income. Finally, hospital networks determine how consumers with different medical needs and geographic access substitute across insurers. A rival whose network attracts healthier consumers affects an insurer differently from one attracting consumers with greater medical needs. Therefore, each insurer's pricing and participation responses will also depend on the availability of rival networks.

Product space Following related models of insurance competition (Dickstein, Ho and Mark, 2024; Tebaldi, 2025; Geddes, 2025), I represent products by insurer, metal tier, and plan type within each county-year, which is the finest choice observed in the enrollment data. All plans in my sample are Health Maintenance Organization plans (HMOs), so I refer to these as insurer–metal products within each county-year. The aggregation retains the dimensions of sorting central to this analysis while abstracting from benefit-design differences within each cell. It also keeps the choice set manageable for repeated pricing and entry calculations. As a result of this aggregation, within firm-metal, the variation in premiums only comes through statutory age rating and

income-based subsidies. Each insurer offers exactly three plans, one in each metal tier: a bronze plan with actuarial value 60%, a silver plan with standard actuarial value 70%, and a gold plan with actuarial value 80%. Consumers with income below 250% FPL qualify for cost-sharing reduction according to a statutory schedule, so actuarial values for the same silver plan offered by the same insurer also vary by income. I assume that all plans offered by the same firm share the same network of hospitals.

Estimation I estimate the demand parameters using observed insurance plan choices. Estimation fits the probability of each individual's choice while matching insurers' enrollment shares within each county–year–risk group. Because individual income is unobserved, I integrate choice probabilities over the assigned distribution of income bins, which determines the premiums and actuarial values each consumer faces.

Let $P_{ij}(b, r_i; \theta, \delta)$ denote the logit probability of choosing product j implied by equation 1, conditional on income bin b and spending-risk group r_i. Write y_i for individual i's observed choice, with $y_i = 0$ for the uninsured, and $q_i(b \mid r_i, Z_i)$ for their assigned income distribution conditional on risk group and observed characteristics. Each individual's contribution to the log pseudo-likelihood is

$$\ell_i(\theta, \delta) = \sum_j \mathbf{1}\{y_i = j\} \log \left[\sum_b q_i(b \mid r_i, Z_i)\, P_{ij}(b, r_i; \theta, \delta) \right],$$

At each candidate vector of preference parameters θ, I recover the fixed effects $\delta(\theta)$ that match observed insurer shares within each county–year–risk group. I then choose θ to maximize the log pseudo-likelihood summed over insured individuals and fractionally weighted uninsured risk rows. During estimation, I take population weights, income and risk assignments, and subsidy eligibility as given. Appendix B.2 details the likelihood, share constraints, and numerical procedure.

Conditional on the maintained income and eligibility assignments, the premium coefficients are identified from variation in net premiums generated by age rating and income-based subsidies. The actuarial-value coefficients are identified from differences across metal tiers and income-dependent cost-sharing reductions within Silver plans. The network-value coefficients are identified from variation in expected network utility across consumers with different demographic

characteristics and geographic locations, holding insurer–county–year–risk fixed effects constant.

Results Table 3 reports the demand estimates and implied economic magnitudes. The net-premium coefficient is more negative for lower-risk consumers, indicating greater sensitivity to out-of-pocket premiums. The implied own-plan net-premium elasticities are -1.56 for lower-risk consumers and -1.72 for higher-risk consumers, holding other plans' premiums and attributes fixed, similar in magnitude to the estimate reported by Abraham et al. (2017). A \$10 increase in monthly net premiums across all plans reduces total enrollment by 3.6 percent, with reductions of 5.65 percent among lower-risk consumers and 1.53 percent among higher-risk consumers. A \$10 increase in monthly gross premiums across all plans, with the benchmark also rising by \$10 and subsidies recalculated, reduces enrollment by 0.33 percent among lower-risk consumers and similarly among higher-risk consumers. In this case, subsidies rise with the benchmark and mute the increase in out-of-pocket premiums for eligible consumers. The actuarial-value coefficients imply monthly willingness to pay of \$29.12 for lower-risk consumers and \$65.50 for higher-risk consumers for a 10-percentage-point increase in actuarial value. For consumers without cost-sharing reductions, this corresponds to moving up one metal tier, holding other plan attributes fixed. These valuations are comparable in magnitude to estimates from other Marketplace studies (Polyakova and Ryan, 2021; Tebaldi, 2025).

To interpret network preferences, I calculate the statewide population average network willingness to pay for each insurer, and compare the networks with the lowest and highest population-average network willingness to pay. Holding other plan attributes fixed, moving between these networks implies average monthly willingness to pay of \$13.74 among lower-risk consumers and \$101.01 among higher-risk consumers, with a population-wide average of \$43 per month. The higher valuation among higher-risk consumers is consistent with Shepard's (2022) finding that sicker consumers are willing to pay more for access to their preferred providers.

Table 3: Demand Estimates and Economic Magnitudes

Attribute	Lower-risk	Higher-risk
Net premium coefficient ($100/month)	−1.478	−1.041
	(0.0174)	(0.0170)
Own-plan net-premium elasticity	−1.56	−1.72
Enrollment change: all net premiums +$10/month	−5.65%	−1.53%
Enrollment change: all gross premiums +$10/month	−0.33%	−0.30%
Monthly WTP for +10 percentage points of actuarial value (e.g., 60% to 70%)	$29.12	$65.50
	($0.51)	($0.89)
Monthly WTP for higher-value hospital network	$13.74	$101.01
	($2.36)	($7.22)

Notes: The table reports the main demand estimates and implied enrollment responses. Standard errors are in parentheses. Lower- and higher-risk groups are defined by the median predicted spending score. Own-plan elasticities proportionally change consumer-specific net premiums for one plan at a time, holding subsidies, other plans' premiums, participation, and plan attributes fixed. Elasticities are averaged using baseline predicted enrollment within each risk group; zero net premiums remain zero. The net-premium experiment adds $10/month to every consumer-specific net premium. The gross-premium experiment adds $10/month to every age-rated gross premium, recalculating subsidies with eligibility and required contributions held fixed. The network comparison uses the insurers with the lowest and highest network values averaged over the same potential-market population pooled across 2017–2019, including counties where they do not currently participate. The estimation sample is a 59,594 person-year random subsample of enrollee-year records across 2017–2019.

4.2 Supply Stage 2: Firm Pricing Problem and Marginal Cost Specification

In this stage, insurers set premiums via a differentiated Bertrand pricing game, having observed each firm's entry decision and marginal cost shocks. Recall that each insurer sells three plans, one in each metal tier, and that networks are taken as given.[12]

Firms choose prices to maximize their profit across plans $j \in \{\text{Bronze, Silver, Gold}\}$, subject to the markup constraint that a firm's premium revenue across plans not exceed 1.25 times the total variable cost:

$$\Pi_{fm}(p) = \sum_{j \in \mathcal{J}_{fm}} \sum_{k} \underbrace{D_{kjfm}(p)}_{\text{Enrollment by type}} \underbrace{\left[p_{kjfm} - MC_{kjfm}\right]}_{\text{Margin on that type}}$$

$$s.t. \quad \text{Premium revenue} \leq 1.25 \times \text{Total variable cost}$$

where consumer types k in each market are characterized by age, sex, income bin, on-exchange indicator, spending risk bin, and network value.[13]

The type-specific marginal cost for each plan has two components and is given by $MC_{kjfm} = HC_{kjfm} + c_{fm}$. HC_{kjfm} is the type-plan-specific expected claims cost. c_{fm} is the residual marginal cost that approximates administrative costs that do not vary by health care utilization, as well as expectational errors not captured by HC_{kjfm}.

Expected claims costs To capture how spending can vary across insurers even for the same individual, I model expected claims costs using both individual and plan characteristics. Specifically, I fit a flexible regression of realized cost on the characteristics that enter demand – spending risk group, demographics, geography – as well as characteristics that are plan-specific, including metal tier, network value, and insurer indicators. Appendix C explains the model estimation and validation in more detail.

Residual marginal costs In standard Bertrand pricing games, marginal costs can be directly inverted from the firms' pricing first-order conditions. However, in my constrained pricing game,

[12] Insurers typically negotiate networks across commercial market segments. Because the individual market is a relatively small segment for most insurers, pricing decisions in this market are unlikely to affect network decisions across segments.

[13] Because federal rules prohibit plan-specific adjustments that reflect risk selection across plans, and the observed price ratio is approximately fixed across metal tiers for the same firm, I model firms as choosing the price of the bronze plan while setting the prices of the silver and gold plans according to observed multipliers between plans.

one cannot infer from the data whether a firm is slack or binding at the observed prices. Therefore, I estimate a slack-MLR candidate that sets the derivative of unconstrained profit with respect to price equal to zero, and a binding-MLR candidate that makes costs equal to 80 percent of premium revenue, then check whether the resulting estimates are consistent with the assumptions under which they were estimated. Additionally, I exclude candidates that imply a zero or negative cost of covering any consumer type under any of the insurer's plans. I also require premium revenue to cover variable costs, maintaining that observed insurers do not operate at a loss before fixed entry costs.[14] I retain an estimate only when these restrictions leave one distinct cost, which occurs in all but six of the year-county-firm cells. In Appendix D, I show that the marginal cost estimates are stable across specifications that include these candidates under various assumptions.

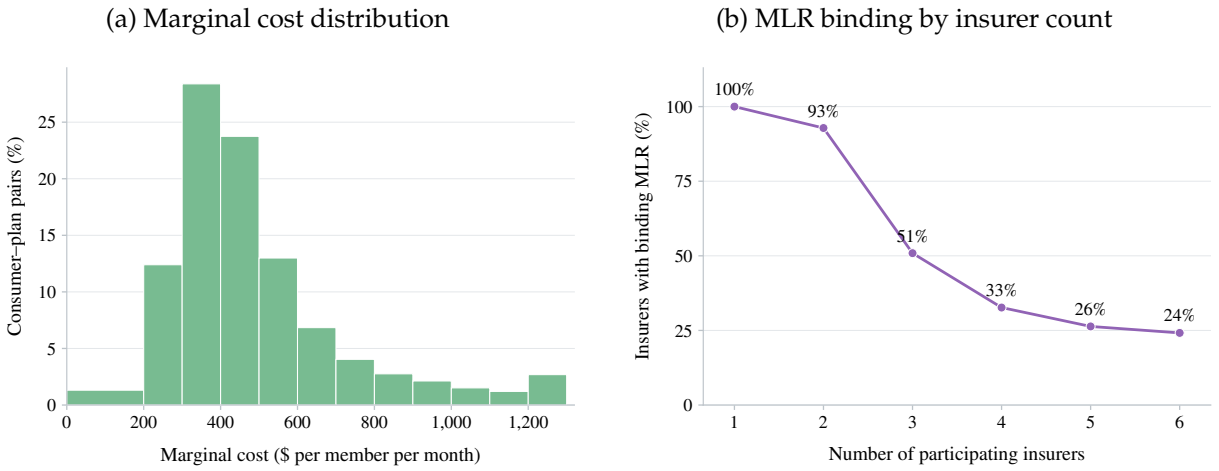
To get residual marginal costs for all potential firm-market cells, I fit a projection model relating the admissible inverted costs to observable firm-market characteristics. The regressors include the firm's lagged presence in the focal county, the share of other counties in the same rating area in which it was active in the previous year, summaries of expected claims and demand utility, and separate firm, county, and year fixed effects. The residuals from this projection form the empirical distribution of ω_{fm}, the marginal cost shock realized after entry but before pricing. See Appendix D for more details.

Results Figure 5 summarizes the estimated marginal costs and the role of MLR regulation in premium setting. Panel (a) plots the distribution of marginal costs across consumers and potential plans, weighting consumer types by their population shares and plans equally. The distribution is right-skewed, with a median of \$429 per member per month. The middle 50 percent of consumer–plan combinations have marginal costs between \$348 and \$563, and the 90th percentile is \$810.

Panel (b) illustrates how MLR regulation restrains pricing power in concentrated markets: fewer competitors give insurers greater scope to raise premiums, but MLR limits further markup. Across all potential entrant sets, the constraint binds in all monopoly observations and gradually binds less often as the number of competitors increases.

[14] Because pricing occurs after marginal cost shocks are realized, the insurer should be able to set the premium implying zero variable profit instead of operating at a loss. Note that nonnegative variable profit is compatible with negative total profit after accounting for entry costs.

Figure 5: Estimated marginal costs and MLR constraints



Notes: Panel (a) pools consumer–plan marginal costs over 2017–2019. Each consumer type receives its population weight, divided equally across its potential plans. Marginal cost includes expected insurer claims and the insurer–county–year residual cost. Bars report the percentage in each cost group; the tails are pooled below $200 and at $1,200 or more. Panel (b) pools simulated baseline pricing equilibria across entrant sets and marginal-cost draws over 2017–2019, giving each participating insurer in each equilibrium equal weight. MLR is classified as binding when the cost-to-revenue ratio is within 0.1 percentage point of 80%. Configurations are not weighted by participation probabilities or enrollment.

4.3 Supply Stage 1: Entry

I model annual insurer participation as a static game of incomplete information (Seim, 2006). The set of potential entrants consists of the six insurers operating in Colorado during my sample period. At the beginning of each period, insurers observe their own fixed costs but know only the distribution of their rivals' fixed costs. Firms form beliefs about the probability each rival enters, and participate when expected variable profits exceed their fixed costs. In equilibrium, insurers' beliefs about their rivals' participation coincide with the participation probabilities implied by those rivals' decisions.

The static formulation is motivated by a focus on local participation decisions by insurers already operating in Colorado. These firms can draw on established administrative infrastructure, while provider relationships supported by other insurance segments can remain available independently of individual-market participation. This reduces the extent to which participating today is necessary to preserve the capacity to serve a market in the future. In this static formulation, fixed costs largely reflect the cost of maintaining provider networks, and I allow fixed costs to differ by prior participation to account for the cost of establishing a local network faced by

insurers without an existing local operation. Together, these features support a static approximation that relates participation to operating profitability while accounting for the cost advantages of existing local infrastructure. Relatedly, the incomplete-information formulation reflects insurers' informational advantage about their own fixed operating costs: competitors can observe existing market presence but may not fully observe the internal resources required to establish or maintain local networks.

Let $\tilde{q}_{jmt}$ be the belief about the probability firm j will enter county m in year t. Firm f's probability of facing rival set $y_{-f,mt}$ is

$$\Pr(y_{-f,mt}|\tilde{q}_{mt}) = \prod_{j \neq f} \tilde{q}_{jmt}^{y_j}(1-\tilde{q}_{jmt})^{1-y_j}$$

Its expected variable profit under that belief is

$$\overline{V}_{fmt}(\tilde{q}_{-f,mt}) = \sum_{y_{-f,mt}} \Pr(y_{-f,mt}|\tilde{q}_{mt}) E_\omega[V_{fmt}(y_{-f,mt}, \omega_{mt})].$$

Under the belief $\tilde{q}_{-f,mt}$, firm f enters when expected variable profit covers its realized fixed cost. Its ex-ante entry probability is

$$q_{fmt}(\tilde{q}_{-f,mt}) = \Pr\left[\overline{V}_{fmt}(\tilde{q}_{-f,mt}) \geq FC_{fmt}(\nu_{fmt})\right]$$

In equilibrium, all firms best respond to the entry probabilities of the other firms, so an equilibrium of this game is characterized by a fixed point in entry probabilities.

I model fixed costs as following a strict half-normal distribution whose scale depends on market size, prior participation, and year:

$$FC_{fmt}(\nu_{fmt}) = \kappa_{fmt}|\nu_{fmt}|, \qquad \nu_{fmt} \sim N(0,1),$$

$$\log \kappa_{fmt} = \mu + \rho \log M_{mt} + \delta L_{fm,t-1} + \lambda_t,$$

where M_{mt} is the market's potential market size measured by the number of potential individual market consumers, $L_{fm,t-1}$ indicates that firm f had at least five enrollees in county m in the

previous year, and λ_t is a year fixed effect common to previously present and absent firms.

Estimation To estimate the fixed-cost parameters, I use a two-stage procedure analogous to Pakes, Ostrovsky and Berry (2007). In the first stage, I estimate the entry policy functions played in the observed equilibrium. In the second stage, I combine the estimated entry probabilities with precomputed pricing stage game profits to construct each firm's expected variable profit from entry, and then estimate the fixed-cost parameters by fitting model-implied entry probabilities to observed entry decisions. This approach avoids repeatedly solving the entry game or imposing an equilibrium-selection rule during estimation.

The main challenge of adapting the Pakes, Ostrovsky and Berry (2007) approach to my setting is that, with consumer selection and imperfect competition, an insurer's profitability depends on a potentially high-dimensional set of own-firm, rival, and market characteristics. The first-stage specification must nevertheless be parsimonious, since including too many predictors relative to the number of observed entry decisions risks overfitting and unstable probability estimates. For these reasons, I first use the solved pricing games to identify characteristics informative about profitability, then select a parsimonious subset based on its ability to predict entry in held-out counties. In the final step in this stage, I use the selected predictors along with the variables that enter the fixed-cost specification to estimate a Logit regression of observed entry.

The fitted probabilities determine the competitive environments insurers expect to face upon entering. For each potential insurer, I calculate its pricing-stage profit conditional on entry under each possible rival configuration, averaged over marginal-cost shock draws, then average across rival configurations using the probabilities implied by the first-stage estimates to obtain expected variable profit from entry.

I estimate the fixed-cost parameters by maximum likelihood using these expected profits and observed participation. In Appendix E.4, I show that estimation using Generalized Method of Moments (GMM) yields similar qualitative results. Appendix E provides the estimation details and validation exercises. Appendix F describes the pricing-game solution procedure.

Results Table 4 reports economic magnitudes implied by the MLE fixed-cost estimates used in the counterfactual calculations. Mean annual fixed cost for an insurer with prior participation is

$232,000 in a 2019 market with 1,214 potential individual-market consumers, the median market size that year. Fixed costs decline less than proportionally as market size falls. The estimated market-size elasticity is 0.436, meaning that a 10% reduction in the number of potential consumers lowers mean fixed costs by 4.5%, holding prior participation and year fixed. Previously present insurers have mean fixed costs 79.7% lower than those of previously absent insurers, conditional on market size and year, consistent with costs of establishing a local operation that continuing insurers have already paid for.

Table 4: Fixed-Cost Estimates and Economic Magnitudes

Statistic	Estimate	95% confidence interval
Mean annual fixed cost at the reference market ($1,000)	232	[169, 301]
Market-size elasticity of mean fixed cost	0.436	[0.267, 0.611]
Fixed-cost reduction associated with prior participation (%)	79.7	[69.5, 86.9]

Notes: The reference is a potential insurer with prior participation in a 2019 market containing 1,214 potential individual-market consumers, the median market size that year. Prior participation means having at least five enrollees in the preceding year. The prior-participation comparison holds market size and year fixed. Confidence intervals are based on 1,000 bootstrap replications.

Model Fit I assess model fit along four dimensions relevant to insurers' responses to Medicaid expansion: market size, expected fixed costs, expected variable profits, and exposure to the eligibility change. Figure 6 compares observed participation with the probabilities implied by the estimated model. Panel (a) shows that the model captures the increase in participation from the smallest third of markets to the largest third of markets, although the participation rates are somewhat underpredicted. Panels (b) and (c) show that the model also reproduces the decline in participation across expected-fixed-cost quartiles and its increase across expected-variable-profit quartiles. Panel (d) assesses fit by expansion exposure, measured as the share of potential consumers in the 138-199% FPL group who would become eligible for Medicaid in the main counterfactual. The model captures the lower participation in more exposed markets. The model closely replicates the participation rate for the middle third of markets ranked by exposure while moderately underpredicting participation rates for the other two groups.

Figure 6: Observed versus fitted participation



Notes: The sample contains 1,098 potential insurer–county–year observations over 2017–2019, with six potential insurers per county-year. Bars show observed participation rates and mean fitted probabilities using the baseline MLE coefficients. Each potential insurer–county–year receives equal weight. Market-size and exposure thirds are defined across county-years; fixed-cost and variable-profit quartiles are defined across potential insurer–county–years. Expected fixed cost in panel (b) is the mean of the estimated half normal distribution. Expansion exposure in panel (d) is the share of potential consumers with assigned incomes of 138–199% FPL.

5 Counterfactual Simulation

I use the estimated demand, pricing, and entry models to simulate an extension of Medicaid eligibility to consumers with incomes between 138 and 199 percent of the federal poverty level, which removes this group from the private individual market. I apply the policy to 2019 markets, the last year in my data. I first describe changes in market outcomes and insurer participation, then compare outcomes across market structures and examine the implications for consumer welfare and government spending across markets of different sizes.

Table 5 reports the effect of extending Medicaid eligibility to adults with incomes between 138 and 199 percent of FPL in 2019. Column 1 holds each insurer's participation probability at its baseline level while allowing premiums, subsidy benchmarks, and consumers' plan choices to adjust. In an average market, the low-risk enrollee share declines by 4.29 percentage points, and average insurer cost rises by $3.27 per enrollee per month. Reflecting the higher insurer costs, gross premiums increase by $15.56, but subsidies rise by $17.11, leaving average net premiums $1.55 below baseline among remaining consumers. Although subsidies largely offset the premium increase, total private enrollment falls by 33.6 percent in the average county as the expansion population leaves the market, and total expected variable profit falls by 33.3 percent. Column 2 reports the full policy response across all markets, additionally allowing insurers' participation probabilities to adjust in equilibrium. Risk composition and insurer costs change relatively little beyond the fixed-participation response, while higher subsidies continue to offset higher gross premiums. Average net premiums remain close to the first-column result, at $1.79 below baseline. As insurer participation declines, consumers face fewer coverage options, and the total enrollment decline increases from 33.6 to 35.0 percent. The decline in total expected variable profit likewise increases from 33.3 to 34.2 percent. The expected number of insurers falls by 7.5 percent, equivalent to an average decline of 0.16 insurers per county.

Columns 3–5 report the full policy response by market size, grouping counties into thirds by their pre-policy potential private-market population and averaging county-level changes within each group. The largest markets experience the greatest decline in the low-risk enrollee share and the largest increase in insurer cost: 5.51 percentage points and $7.47 per enrollee per month, respectively, compared with 3.87 percentage points and $2.53 in the smallest markets. Subsidies off-

set higher gross premiums in every group, leaving average net premiums below baseline throughout. Total private enrollment falls by 39.5 percent in the smallest markets, compared with 33.0 percent in the middle group and 32.5 percent in the largest. Total expected variable profit follows a similar pattern, declining by 39.2, 32.5, and 31.0 percent, respectively. Expected insurer counts fall by 9.9 percent in the smallest markets, compared with 7.3 percent in the middle group and 5.6 percent in the largest. These correspond to similar absolute declines of 0.16, 0.14, and 0.16 insurers per county, reflecting the fact that insurer participation is more sparse in the smallest markets at baseline.

Table 5: Effects of extending Medicaid eligibility

	Fixed insurer participation	Full response w/ endogenous insurer participation			
Outcome	All markets (1)	All markets (2)	Smallest third (3)	Middle third (4)	Largest third (5)
Panel A. Risk composition and costs					
Low-risk enrollee share (pp change)	−4.29	−4.38	−3.87	−3.71	−5.51
Insurer cost ($ per enrollee-month)	+3.27	+4.32	+2.53	+2.80	+7.47
Panel B. Premiums and subsidies					
Gross premium ($ per enrollee-month)	+15.56	+17.59	+18.08	+20.55	+14.32
Premium subsidy ($ per enrollee-month)	+17.11	+19.39	+19.14	+22.14	+17.00
Net premium ($ per enrollee-month)	−1.55	−1.79	−1.05	−1.59	−2.69
Panel C. Enrollment, profitability, and participation					
Total private enrollment (% change)	−33.63	−34.99	−39.55	−33.01	−32.55
Total expected variable profit (% change)	−33.28	−34.19	−39.21	−32.54	−30.98
Expected insurer count (% change)	–	−7.54	−9.88	−7.29	−5.55
Expected insurer count (absolute change)	–	−0.16	−0.16	−0.14	−0.16

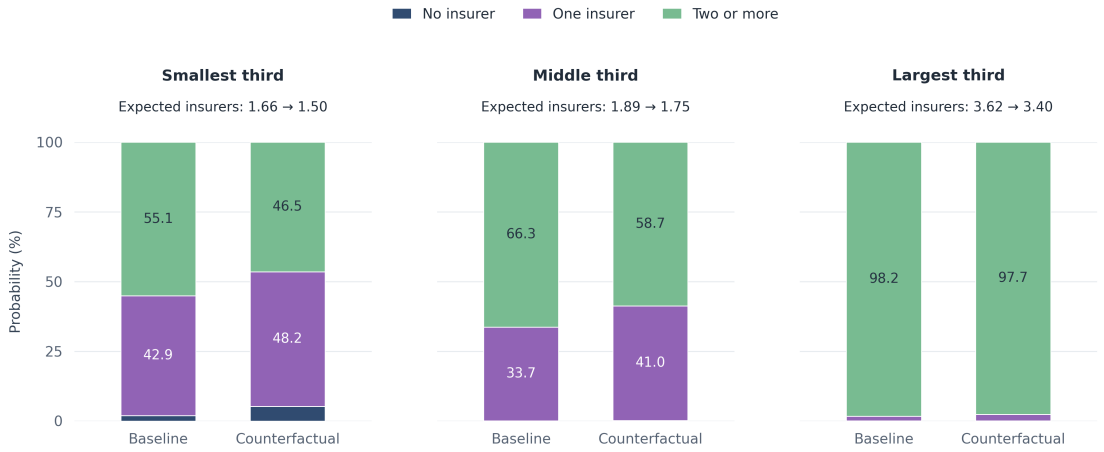
Notes: Markets are grouped into thirds by their baseline potential population. Changes are calculated within each county and then averaged across counties, giving each county equal weight. Enrollment, risk composition, insurer costs, and profits are compared with their pre-policy values. Premium and subsidy changes compare the same population before and after the policy, excluding the group receiving extended Medicaid eligibility. Total expected variable profit sums the profits insurers expect to earn if they were to participate, weighting each insurer's contribution by its participation probability. The expected insurer count is the sum of these participation probabilities.

The similar decline in expected insurer counts has more pronounced consequences in smaller markets. Figure 7 divides markets into thirds, ranked by the size of the potential private market population pre-expansion, and shows that the participation response shifts smaller markets more strongly toward monopoly and no coverage.[15] While expected participation falls by sim-

[15]Regulators may intervene to prevent markets from being left without an insurer. The simulated no-insurer out-

ilar amounts across market-size groups, because smaller markets have fewer insurers at baseline, participation declines in smaller markets translate into larger increases in the probabilities of monopoly and no coverage. In the smallest markets, the probability of no insurer rises from 2.0 to 5.2 percent, and monopoly probability rises from 42.9 to 48.2 percent. In middle-sized markets, monopoly probability also rises from 33.7 to 41.0 percent. The largest markets instead retain 3.40 expected insurers and a 97.7 percent probability of multiple insurers.

Figure 7: Changes in insurer participation by market size



Notes: Markets are grouped by the original pre-expansion potential private market population. Expected insurer counts sum the potential firms' entry probabilities within each county and are then averaged within each group using remaining potential population weights. Each pair of bars compares baseline and counterfactual probabilities of zero, one, and two or more insurers. Within each market, entrant set probabilities are summed to obtain probabilities of insurer counts, then averaged within each group weighted by the remaining potential population.

The decline in insurer participation matters because different market structures imply different enrollment, consumer surplus, and subsidy spending. Table 6 compares these outcomes at counterfactual premiums and subsidy benchmarks in equilibria with different numbers of entrants. Compared with configurations containing multiple insurers, monopoly configurations have a 16.7 percentage-point lower enrollment rate and 4.1% higher markup on average. Consumer surplus per potential consumer per month is also lower by $65.95 (46%) under monopoly. Although subsidies per enrollee are higher under monopoly, spending per potential consumer

comes therefore indicate pressure on coverage availability that could require an additional policy response, potentially involving financial support or regulatory concessions. Even when intervention preserves coverage, ensuring that one insurer remains does not restore lost competition or coverage options, so declining participation can still have consequences for consumers.

falls from $444.25 with multiple insurers to $340.92 with one insurer, reflecting lower enrollment.

Table 6: Outcomes across participation configurations

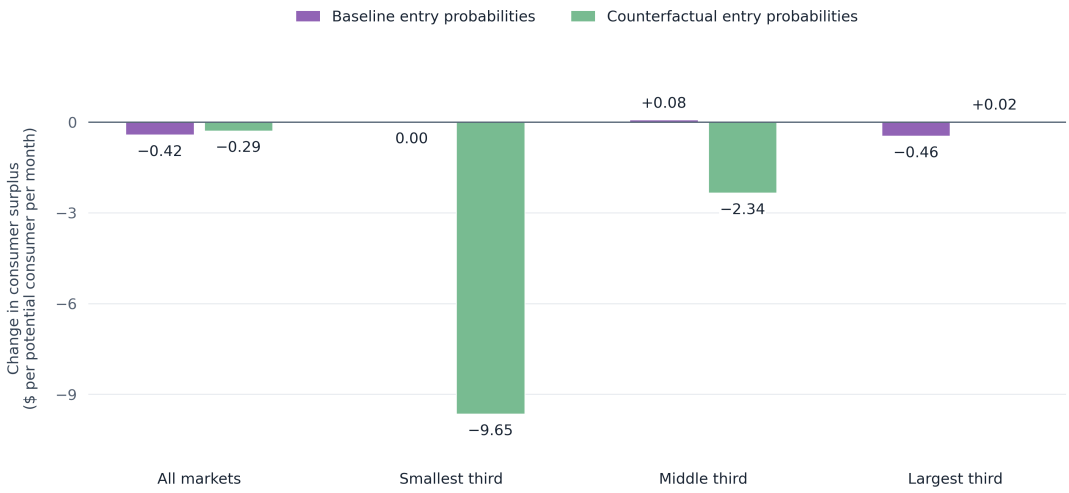
Outcome	One insurer	2+ insurers
Enrollment rate (%)	37.42	54.09
Average markup (%)	25.00	24.01
Consumer surplus ($ per potential consumer per month)	78.30	144.25
Government subsidy ($ per enrollee per month)	904.93	812.23
Government subsidy ($ per potential consumer per month)	340.92	444.25

Notes: This table compares market outcomes under monopoly and multiple-insurer configurations. All outcomes use counterfactual prices and subsidy benchmarks. Government subsidy spending refers to spending towards premium tax credits and excludes Medicaid spending. Market outcomes are calculated within each equilibrium, averaged equally over equilibria within each column. Within each equilibrium, enrollment rates, consumer surplus per potential consumer, and government subsidy per potential consumer use the remaining potential population, including uninsured consumers, as the denominator; subsidy per enrollee uses private-market enrollment. Markup is the enrollment-weighted average of insurers' percentage markups within each equilibrium. Each insurer's markup is premium revenue minus variable cost, divided by variable cost and expressed in percent.

The preceding analysis suggests that subsidies limit the effect of premium changes on consumers with participation fixed, while the greater shift toward monopoly in smaller markets could reduce consumer surplus more sharply in these markets. Consistent with these mechanisms, Figure 8 shows modest welfare changes when participation is held fixed and larger losses in smaller markets when participation adjusts. At baseline participation probabilities, consumer surplus falls by $0.42 per remaining potential consumer per month overall, with small differences across market-size groups. Allowing participation to adjust produces losses of $9.65 in the smallest markets and $2.34 in the middle group, alongside a small gain of $0.02 in the largest. The overall change is a loss of $0.29 per remaining potential consumer per month.

Figure 9 illustrates how changes in monthly subsidies per enrollee combine with enrollment responses to determine monthly subsidy spending per remaining potential consumer. With participation held at baseline (Panel A), subsidies per enrollee rise in every market-size group, consistent with subsidies offsetting higher gross premiums. Spending per potential consumer also increases throughout, with statewide increases of $19.08 per enrollee and $7.87 per potential consumer. Allowing participation to adjust (Panel B) further increases subsidies per enrollee in every group, but produces different spending responses across markets. In the smallest markets, subsidies per enrollee rise by $22.86, yet spending per potential consumer falls by $6.72 relative to baseline, compared with a $12.23 increase under fixed participation. This suggests that the enrollment

Figure 8: Consumer-surplus changes overall and by market size

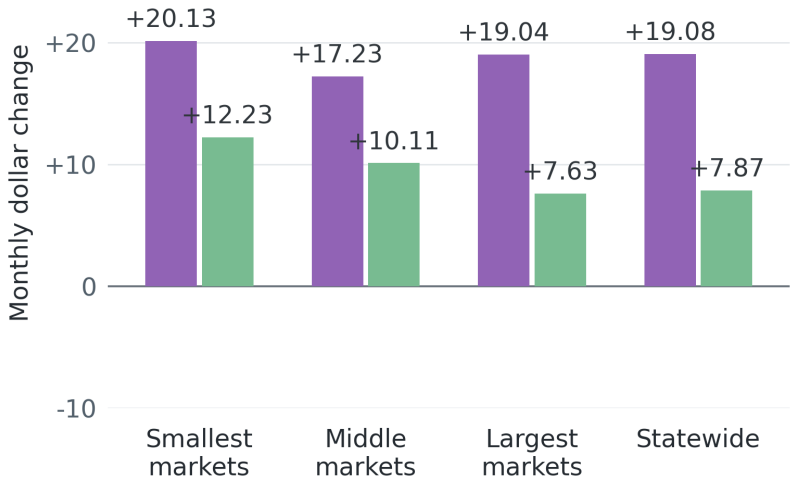


Notes: Bars report changes from baseline consumer surplus in monthly dollars per remaining potential consumer. Both counterfactual series use counterfactual premiums and subsidy benchmarks; the left bar retains baseline entry probabilities and the right bar uses counterfactual entry probabilities. Baseline consumer surplus uses baseline prices and entry probabilities for the same remaining population. Outcomes are averaged over cost draws and then over entrant sets with the indicated probabilities. Group changes equal summed consumer-surplus changes divided by the group's remaining population.

decline attributable to participation responses more than offsets the higher subsidy per enrollee in these markets. In middle-sized markets, participation adjustment similarly moderates the spending increase, from \$10.11 to \$8.13 per potential consumer. In the largest markets, however, both enrollment and subsidies per enrollee rise relative to fixed participation, increasing spending from \$7.63 to \$9.89 per potential consumer above baseline. In these markets, reduced participation by low-premium insurers helps shift probability toward equilibria with higher subsidy benchmarks, raising subsidies per enrollee enough to lower net premiums on remaining plans and encourage enrollment.

Figure 9: Government premium subsidy spending changes overall and by market size

Panel A. Baseline participation



Panel B. Counterfactual participation



Notes: Bars report monthly dollar changes relative to baseline prices and participation for the same remaining population. Both panels use counterfactual premiums and subsidy benchmarks; Panel A uses baseline participation probabilities and Panel B uses counterfactual participation probabilities. Outcomes are averaged over cost draws and entrant sets with the indicated probabilities. Within each market-size group, subsidy per enrollee divides total subsidy payments by total private-market enrollment, while subsidy per potential consumer divides total payments by the remaining potential population, including uninsured consumers. Changes are differences between these ratios in each scenario and the baseline. Government subsidy spending covers premium tax credits and excludes Medicaid spending.

A back-of-the-envelope calculation places these subsidy changes in the context of the policy's overall fiscal effect. Using Colorado's FY2019 Medicaid benefit spending of $4,349 per full-year-equivalent enrollee among adults made eligible by the ACA expansion to 138 percent of FPL as a cost benchmark (Medicaid and CHIP Payment and Access Commission, 2020) and assuming that the entire expansion population—those with incomes of 138–199 percent of FPL—enrolls in Medicaid, I estimate that covering this group would cost about $301 million annually.[16] The policy also eliminates approximately $471 million in annual premium subsidies for the expansion group, while increasing subsidies for remaining private-market consumers by $30 million. Together, these changes imply approximately $140 million in annual savings across federal and state governments.

6 Conclusion

This paper studies the consequences of expanding public insurance for the private individual market. The analysis highlights insurer participation as a central determinant of the policy's welfare effects, reflecting two features of the market. First, price-linked subsidies dampen the effect of premium changes on consumers, limiting the welfare consequences of insurers' pricing responses. Second, consumer welfare is substantially lower under monopoly or no coverage than when multiple insurers compete. The expansion sharply increases the likelihood of these outcomes, making the loss of private insurance options consequential for consumers who remain outside the public program.

Although this analysis treats income as fixed, households' incomes fluctuate, and broader Medicaid eligibility can reduce the need to switch between Medicaid and other forms of coverage as earnings change. Preserving continuity of coverage provides an important rationale for expansion, motivating further research on the value of Medicaid under income uncertainty. Expansion may also affect the continued availability of hospital care. By covering previously uninsured patients, it can reduce uncompensated care, strengthen hospitals' finances, and reduce the risk of closure (Dranove, Garthwaite and Ody, 2016; Lindrooth et al., 2018). An expansion could therefore discourage private insurer participation while helping sustain hospitals, creating a potential

[16]The assumed full Medicaid take-up among the expansion population provides a lower bound on the net decrease in public spending. Lower take-up would reduce Medicaid spending further and result in larger net savings.

trade-off between insurance availability and provider availability. Hospital availability remained stable in Colorado during my sample period, but studying settings in which Medicaid plays a larger role in hospital viability would allow future research to assess how the benefits of preserving access to hospital care compare with the losses from reduced private insurance choice.

References

Abraham, Jean, Coleman Drake, Daniel W. Sacks, and Kosali I. Simon. 2017. "Demand for Health Insurance Marketplace Plans Was Highly Elastic in 2014–2015." NBER Working Paper 23597. https://www.nber.org/papers/w23597.

Agency for Healthcare Research and Quality. n.d. "Medical Expenditure Panel Survey: Download Data Files." *https://meps.ahrq.gov/data_stats/download_data_files.jsp*, Accessed October 28, 2026.

Center for Improving Value in Health Care. 2026. "Commercial Data in the CO APCD." *https://civhc.org/news/commercial-data-in-the-co-apcd/*, September 22.

Centers for Medicare & Medicaid Services. 2013. "Weekly Assister Newsletter." *https://content.govdelivery.com/accounts/USCMS/bulletins/9b884c*, December 24.

Centers for Medicare & Medicaid Services. 2026. "December 2025: Medicaid and CHIP Eligibility Operations and Enrollment Snapshot." *https://www.medicaid.gov/resources-for-states/downloads/eligib-oper-and-enrol-snap-dec2025.pdf*, March 27.

Centers for Medicare & Medicaid Services. n.d.*a*. "2019 Provider of Services (POS) File." *https://www.hhs.gov/guidance/document/2019-provider-services-pos-file*, Accessed October 28, 2026.

Centers for Medicare & Medicaid Services. n.d.*b*. "Medical Loss Ratio Data and System Resources." *https://www.cms.gov/marketplace/resources/data/medical-loss-ratio-data-systems-resources*, Public use files for reporting years 2017–2019, released December 5, 2018; October 1, 2019; and October 20, 2020, respectively. Accessed October 28, 2026.

Centers for Medicare & Medicaid Services. n.d.*c*. "Medical Loss Ratio: Getting Your Money's Worth on Health Insurance." *https://www.cms.gov/CCIIO/Resources/Fact-Sheets-and-FAQs/medical-loss-ratio*, Accessed October 28, 2026.

Clemens, Jeffrey. 2015. "Regulatory Redistribution in the Market for Health Insurance." *American Economic Journal: Applied Economics*, 7(2): 109–134.

Colorado Division of Insurance. 2018. "Health Insurance Cost Report: Calendar Year 2017." Colorado Department of Regulatory Agencies, Division of Insurance. Published December 10, 2018; corrected March 6, 2019.

Colorado General Assembly. 2026. "SB26-178: Health Insurance Affordability Measures." *https://leg.colorado.gov/bill_files/116949/download*, Signed June 2, 2026.

Colorado Health Institute. 2021. "Eligible but Not Enrolled 2019: Continuing an Upward Trend—Methodology." Colorado Health Institute.

Connect for Health Colorado. 2019. "By the Numbers: Open Enrollment Report, Plan Year 2019." *https://connectforhealthco.com/c4-media/wp-content/uploads/2019/04/OpenEnrollmentReport.pdf*, Accessed October 28, 2026.

Corlette, Sabrina, Jason Levitis, Erik Wengle, and Rachel Swindle. 2023. "The Basic Health Program: Considerations for States and Lessons from New York and Minnesota." Urban Institute. April 26. https://www.urban.org/research/publication/basic-health-program.

Cutler, David M., and Jonathan Gruber. 1996. "Does Public Insurance Crowd out Private Insurance?" *Quarterly Journal of Economics*, 111(2): 391–430.

Diamond, Rebecca, Michael J. Dickstein, Timothy McQuade, and Petra Persson. 2026. "Insurance without Commitment: Evidence from the ACA Marketplaces." Working paper, June 13, 2026. https://pages.stern.nyu.edu/~mjd19/papers/insurance_wo_commitment_ddmp.pdf.

Dickstein, Michael J., Kate Ho, and Nathaniel Mark. 2024. "Market segmentation and competition in health insurance." *Journal of Political Economy*, 132(1): 96–148.

Dorn, Stan, and Jennifer Tolbert. 2014. "The ACA's Basic Health Program Option: Federal Requirements and State Trade-Offs." KFF. November 25. https://www.kff.org/affordable-care-act/the-acas-basic-health-program-option-federal-requirements-and-state-trade-offs/.

Dranove, David, Craig Garthwaite, and Christopher Ody. 2016. "Uncompensated Care Decreased at Hospitals in Medicaid Expansion States but Not at Hospitals in Nonexpansion States." *Health Affairs*, 35(8): 1471–1479.

Ericson, Keith M. Marzilli, and Amanda Starc. 2015. "Pricing Regulation and Imperfect Competition on the Massachusetts Health Insurance Exchange." *Review of Economics and Statistics*, 97(3): 667–682.

Fischer, Andreas. 1992. "A Special Newton-Type Optimization Method." *Optimization*, 24(3–4): 269–284.

Fleitas, Sebastián, Caitlyn Fleming, Gautam Gowrisankaran, and Anthony T. Lo Sasso. 2025. "Group Health Insurance Versus ACA Marketplaces: Selection, Subsidies, and Welfare." NBER Working Paper 33938. Issued June 2025; revised February 2026. https://www.nber.org/papers/w33938.

Geddes, Eilidh. 2025. "The Effects of Price Regulation in Markets with Strategic Entry: Evidence from Health Insurance Markets." Working paper, May 2025. https://eilidhgeddes.com/publication/rating_areas/.

Group, Julia. 2019. "Regional Accountable Entities." Colorado Legislative Council Staff 19-20. November. https://content.leg.colorado.gov/sites/default/files/rae_issue_brief.pdf.

Gruber, Jonathan, and Kosali Simon. 2008. "Crowd-out 10 Years Later: Have Recent Public Insurance Expansions Crowded out Private Health Insurance?" *Journal of Health Economics*, 27(2): 201–217.

Ho, Kate, and Robin S. Lee. 2017. "Insurer competition in health care markets." *Econometrica*, 85(2): 379–417.

Ho, Katherine. 2006. "The welfare effects of restricted hospital choice in the US medical care market." *Journal of Applied Econometrics*, 21(7): 1039–1079.

Internal Revenue Service. 2016. "Revenue Procedure 2016-24." *https://www.irs.gov/pub/irs-drop/rp-16-24.pdf*.

Internal Revenue Service. 2017. "Revenue Procedure 2017-36." *https://www.irs.gov/pub/irs-drop/rp-17-36.pdf*.

Internal Revenue Service. 2018. "Revenue Procedure 2018-34." *https://www.irs.gov/pub/irs-drop/rp-18-34.pdf*.

Jaffe, Sonia, and Mark Shepard. 2020. "Price-Linked Subsidies and Imperfect Competition in Health Insurance." *American Economic Journal: Economic Policy*, 12(3): 279–311.

Kanimian, Salpy. 2025. "Spatial Disparities, Selection, and Segmentation in Health Insurance." Job market paper, December 2025. https://www.salpykanimian.com/.

KFF. 2025. "How Individual Market Enrollment Changed with the Enhanced Premium Tax Credits." *https://www.kff.org/affordable-care-act/how-individual-market-enrollment-changed-with-the-enhanced-premium-tax-credits/*, June 12; corrected June 18. First-quarter 2025 enrollment estimate.

Lindrooth, Richard C., Marcelo C. Perraillon, Rose Y. Hardy, and Gregory J. Tung. 2018. "Understanding the Relationship Between Medicaid Expansions and Hospital Closures." *Health Affairs*, 37(1): 111–120.

Mahoney, Neale, and E. Glen Weyl. 2017. "Imperfect Competition in Selection Markets." *Review of Economics and Statistics*, 99(4): 637–651.

Medicaid and CHIP Payment and Access Commission. 2020. "MACStats: Medicaid and CHIP Data Book." Medicaid and CHIP Payment and Access Commission.

Oregon Health Authority. 2024. "Oregon Health Plan Bridge: Frequently Asked Questions." *https://www.oregon.gov/oha/HSD/OHP/Tools/Bridge-FAQ-EN.pdf*, May 3.

Pakes, Ariel, Michael Ostrovsky, and Steven Berry. 2007. "Simple Estimators for the Parameters of Discrete Dynamic Games (with Entry/Exit Examples)." *RAND Journal of Economics*, 38(2): 373–399.

Peng, Lizhong. 2017. "How Does Medicaid Expansion Affect Premiums in the Health Insurance Marketplaces? New Evidence from Late Adoption in Pennsylvania and Indiana." *American Journal of Health Economics*, 3(4): 550–576.

Polyakova, Maria, and Stephen P. Ryan. 2021. "Subsidy Targeting with Market Power." NBER Working Paper 26367. Originally issued October 2019; revised August 2021. https://www.nber.org/papers/w26367.

Seim, Katja. 2006. "An empirical model of firm entry with endogenous product-type choices." *The RAND Journal of Economics*, 37(3): 619–640.

Sen, Aditi P., and Thomas DeLeire. 2018. "How Does Expansion of Public Health Insurance Affect Risk Pools and Premiums in the Market for Private Health Insurance? Evidence from Medicaid and the Affordable Care Act Marketplaces." *Health Economics*, 27(12): 1877–1903.

Shepard, Mark. 2022. "Hospital network competition and adverse selection: evidence from the Massachusetts health insurance exchange." *American Economic Review*, 112(2): 578–615.

Starc, Amanda. 2014. "Insurer Pricing and Consumer Welfare: Evidence from Medigap." *RAND Journal of Economics*, 45(1): 198–220.

Tebaldi, Pietro. 2025. "Estimating Equilibrium in Health Insurance Exchanges: Price Competition and Subsidy Design under the ACA." *Review of Economic Studies*, 92(1): 586–620.

U.S. Census Bureau. n.d.*a*. "2019 ACS Summary File Data and Documentation." *https://www.census.gov/programs-surveys/acs/data/summary-file/sequence-based/2019.html*, Accessed October 28, 2026.

U.S. Census Bureau. n.d.*b*. "American Community Survey: Public Use Microdata Sample Documentation." *https://www.census.gov/programs-surveys/acs/microdata/documentation.html*, Accessed October 28, 2026.

U.S. Census Bureau. n.d.*c*. "Current Population Survey: Annual Social and Economic Supplement Datasets." *https://www.census.gov/data/datasets/time-series/demo/cps/cps-asec.html*, Accessed October 28, 2026.

U.S. Department of Health and Human Services. 2025. "45 CFR 156.420: Plan Variations." *https://www.govinfo.gov/content/pkg/CFR-2025-title45-vol2/pdf/CFR-2025-title45-vol2-sec156-420.pdf*, Code of Federal Regulations, October 1 edition.

U.S. Department of Health and Human Services. n.d. "45 CFR 156.200: QHP Issuer Participation Standards." *https://www.ecfr.gov/current/title-45/subtitle-A/subchapter-B/part-156/subpart-C/section-156.200*, Accessed October 28, 2026.

Zahn, Matthew V. 2025. "Entry and Competition in Insurance Markets: Evidence from Medicare Advantage." Working paper, March 21, 2025. https://matthew-zahn.github.io/files/papers/choice_cost/JMPZahn.pdf.

A Details on Income Distribution Assignment

This appendix describes how I combine administrative enrollment records with external data to assign income probabilities. Each insured member-year receives a probability distribution over six income categories: 0–137, 138–199, 200–249, 250–299, 300–399, and 400 percent or more of the federal poverty level (FPL). Simulated uninsured individuals are assigned a joint probability distribution over the six income categories and two predicted spending-risk groups.

For each insured enrollee, the assignment combines coverage-specific income priors, geographic adjustments, and predicted spending risk. I start with a statewide income distribution specific to the enrollee's exchange status and year and adjust it using the income distribution in the enrollee's county and age group. Next, I use MEPS to estimate a joint distribution between income, broad age group, sex, and spending risk, applying the scoring rule and cutoff in Appendix C.1.

Statewide income distribution For on-exchange enrollees, I use the reported FPL distribution from the Connect for Health Colorado (Colorado's state-run ACA exchange) enrollment reports. For off-exchange enrollees, I use pooled 2017–2019 Colorado observations from the CPS ASEC (U.S. Census Bureau, n.d.*c*) and calculate the statewide income distribution among adults between 18 and 64 with direct-purchase, non-Marketplace coverage. For the uninsured sample, I use the yearly ACS Public Use Microdata Sample (PUMS) (U.S. Census Bureau, n.d.*b*) to calculate the statewide income distribution among adults between 18 and 64 who report being uninsured.

ACS geographic adjustment The second step uses the ACS to construct a county-by-age adjustment. This step is applied to all three coverage groups. For each county, age group (18–24, 25–34, 35–44, 45–54, and 55–64), and income category, I calculate the category's share of the county-age population and divide it by the category's share of the statewide adult population. This ratio measures whether the income category is relatively more or less prevalent in that county-age cell than in Colorado overall.[17]

[17] The published ACS categories do not align exactly with all six analysis categories. The ACS reports 125–149 percent FPL as one category. I divide this count equally between the 0–137 and 138–199 percent categories. Similarly, I divide the published 200–299 percent category equally between the 200–249 and 250–299 percent categories.

MEPS joint income–spending-risk distribution The MEPS provides information on the relationship between income and spending risk, which are not jointly observed in the administrative data. I pool West-region MEPS respondents from 2017–2019 and classify their spending risk using the same demographic and diagnosis-based scoring rule and cutoff applied to APCD enrollees, as described in Appendix C. I estimate separate relationships for insured and uninsured adults, pooling on- and off-exchange enrollees within the insured group.

Within each coverage group, I calculate the probability of low and high spending risk conditional on income, age, and sex. To limit sparse cells, I use three income groups—0–137, 138–199, and 200 percent or more of FPL—and two age groups, 18–44 and 45–64. The four analysis income categories at or above 200 percent FPL therefore share the same conditional risk probabilities within an age–sex group. When either risk group is absent from an age–sex–income cell, I pool sexes within the corresponding age–income cell.

For the uninsured group, I adjust the MEPS-based risk probabilities using a prevalence ratio to account for differences between self-reported health information and administrative claims, as well as differences between Colorado and the broader Western region. Specifically, I calculate the ratio of insured high-risk prevalence in the APCD to age–sex-standardized insured high-risk prevalence in MEPS, then multiply each MEPS high-risk probability for uninsured adults by this ratio and set the low-risk probability to its complement. This adjustment assumes that the proportional discrepancy observed among insured adults also applies to uninsured adults.

I combine these conditional risk probabilities with the coverage-specific income probabilities constructed above. Let $q_g(b \mid Z)$ denote the income probability for coverage group (g), income category b, and characteristics Z derived from the first two steps, and let $L_g(r \mid b, Z)$ denote the corresponding MEPS-based probability of risk group r, including the prevalence adjustment for uninsured adults. The coverage-specific joint weights are

$$J_g(b, r \mid Z) = q_g(b \mid Z) L_g(r \mid b, Z).$$

Appendix B.2 describes how these inputs are combined into the risk-conditional income probabilities used in insurance demand estimation.

B Demand Estimation Details

B.1 Hospital choice model and estimation

Hospital-choice data The hospital-choice model uses inpatient episodes observed in the Colorado All-Payer Claims Database from 2017 through 2019. The sample is restricted to individual-market enrollees between ages 18 and 64 with a qualifying inpatient admission.[18]

Each admission is assigned to one of eight broad diagnosis groups. Starting with the MS-DRG/MDC-based classification of admissions into broad diagnosis groups, I then separately use the principal diagnosis code to identify admissions associated with cancer, allowing cancer cases that span multiple MDCs to be classified into a common cancer category. For each admission, the choice set consists of Colorado hospitals classified as short-term or critical access with valid distance information for the patient's ZIP code. Hospital-year alternatives that are never chosen in the estimation sample are removed.

For each hospital, I construct measures of service availability in five areas using the CMS Provider of Services files (Centers for Medicare & Medicaid Services, n.d.*a*): cardiac care, maternity and neonatal care, mental-health and substance-use treatment, advanced imaging, and oncology. Each measure counts the relevant services available at the hospital. For example, a hospital with a catheterization laboratory, a coronary-care unit, and open-heart surgery is counted as having a cardiac measure of three.

Hospital-choice model Conditional on an inpatient admission and diagnosis group d, individual i, enrolled with insurer f, chooses hospital h in year t. Latent utility is

$$u^H_{ifdht} = \eta_{ht} + \lambda^D_{g(i)} \text{Distance}_{ih} + \lambda^N_d \text{InNetwork}_{fht} + \sum_{s \in \mathcal{S}} \lambda^S_s \mathbf{1}\{d = d(s)\} \text{Service}_{hs} + \nu_{ifdht}, \quad (2)$$

where η_{ht} is a hospital-year fixed effect, $g(i)$ is the hospital-referral region (HRR) containing the individual's ZIP code, and Distance_{ih} is distance in kilometers. The in-network coefficient varies across the eight diagnosis groups. The set $\mathcal{S}$ contains cardiac, maternity, mental-health, imaging, and oncology services; each service measure enters only for its corresponding diagnosis group.

[18] Claims belonging to overlapping inpatient stays are consolidated into episodes. Episodes associated with more than one payer are excluded.

ν_{ifdht} is the i.i.d. logit error. I estimate the model by maximum likelihood.

Table 7 reports the resulting estimates. As expected, distance reduces hospital utility in every referral region, with an additional kilometer lowering utility by between 0.022 and 0.083 units. In-network status raises utility in each of the diagnosis groups, although the estimate for mental and behavioral diagnosis is small and imprecise. The positive diagnosis-service interactions indicate that patients place greater value on hospitals offering services relevant to their admission.

Table 7: Hospital-choice estimates used to construct network value

Variable	Estimate	Standard error
Distance by hospital-referral region		
Distance × Boulder	-0.083	0.004
Distance × Colorado Springs	-0.040	0.002
Distance × Denver	-0.066	0.002
Distance × Fort Collins	-0.052	0.003
Distance × Grand Junction	-0.030	0.001
Distance × Greeley	-0.066	0.003
Distance × Other	-0.022	0.001
In-network status by diagnosis group		
In network × Circulatory and respiratory	1.659	0.153
In network × Digestive	2.211	0.156
In network × Musculoskeletal	1.954	0.170
In network × Neurological and sensory	1.137	0.209
In network × Reproductive, maternal, and neonatal	2.147	0.131
In network × Mental and behavioral	0.121	0.263
In network × Cancer	2.957	0.382
In network × Other	1.537	0.157
Diagnosis-specific hospital services		
Circulatory and respiratory × cardiac services	0.182	0.071
Reproductive, maternal, and neonatal × maternity services	0.101	0.041
Mental and behavioral × mental-health services	0.936	0.140
Cancer × imaging services	0.471	0.145
Cancer × oncology services	0.377	0.192
Hospital-year FE		Yes
Number of episodes		15,021

Notes: This table reports the estimates of the hospital demand model in equation (2). The estimation sample consists of inpatient episodes of Colorado individual market enrollees from 2017–2019.

Expected network value. The insurance-demand model values access to hospital care before diagnoses are realized. Let v^H_{ifdht} denote the deterministic component of hospital utility in equation (2). The logit model gives the expected value of hospital choice conditional on diagnosis as a

log sum. I weight this value by the probability of an admission in diagnosis group d:

$$NV_{ift} = \sum_d \gamma_i^d \log\left[\sum_{h \in \mathcal{H}_{it}} \exp(v_{ifdht}^H)\right].$$

The weights combine admission rates with diagnosis shares conditional on admission, estimated by sex and age group (18–34, 35–50, and 51–64). Hospitals outside an insurer's network remain in the hospital choice set, with their in-network indicator set to zero. For uninsured consumers, that indicator is zero for every hospital. Thus, their outside option retains access to hospital care. The additive extreme-value constant cancels when comparing insurers and uninsured access. This construction allows network values to differ with medical needs and geography, while the insurance-demand coefficients separately allow the valuation of those differences to vary with spending risk.

B.2 Insurance choice and constrained estimation

For each consumer and risk group, I integrate the logit choice probabilities over assigned income.[19] Let $v_{ij}(b, r; \theta, \delta)$ be deterministic insurance utility from equation 1, with utilities expressed relative to the uninsured alternative. Conditional choice probabilities are

$$P_{ij}(b, r; \theta, \delta) = \frac{\exp[v_{ij}(b, r; \theta, \delta)]}{\sum_{k \in \mathcal{J}_i \cup \{0\}} \exp[v_{ik}(b, r; \theta, \delta)]},$$

where 0 denotes being uninsured. Index each observation by i, let w_i be their population mass, and let y_i be their observed choice. The weighted log pseudo-likelihood is

$$\mathcal{L}(\theta, \delta) = \sum_i w_i \sum_{j \in \mathcal{J}_i \cup \{0\}} \mathbf{1}\{y_i = j\} \log\left[\sum_b q_i(b \mid r_i, Z_i) P_{ij}(b, r_i; \theta, \delta)\right].$$

All income probabilities, risk weights, and subsidy eligibility assignments are held fixed during estimation. To construct $q_i(b \mid r_i, Z_i)$, I marginalize over coverage channels by averaging the joint income–risk distributions from Appendix A using the channel-specific population shares conditional on Z_i, then normalize across income categories within each risk group. Therefore, consumers with the same r_i, Z_i receive the same income probabilities regardless of realized coverage.

[19] The assigned income probabilities and fractional uninsured risk weights follow Appendix A.

This avoids conditioning the income weights on the coverage outcome explained by the demand model.

The constraints require predicted insurer shares to match observed shares within each county–year–risk cell. Let $\mathcal{I}_{mtr}$ denote the rows in such a cell, $W_{mtr} = \sum_{i \in \mathcal{I}_{mtr}} w_i$, and s^{obs}_{fmtr} the observed insurer share, including uninsured consumers in the denominator. For every insurer with positive observed share, I impose

$$\frac{1}{W_{mtr}} \sum_{i \in \mathcal{I}_{mtr}} w_i \sum_b q_i(b \mid r, Z_i) \sum_{j : f(j) = f} P_{ij}(b, r; \theta, \delta) = s^{\text{obs}}_{fmtr}.$$

At each candidate θ, a share contraction recovers the fixed effects $\delta(\theta)$. The outer optimization maximizes $\mathcal{L}[\theta, \delta(\theta)]$. The outside-option fixed effect is normalized to zero.

Counterfactual entry requires demand fixed effects for all potential insurers, including combinations of insurer, county, year, and risk group without a finite fixed effect identified by observed enrollment. I project the estimated fixed effects on observable characteristics to estimate the remaining values. The projection includes year, insurer, and county-group fixed effects; the high-spending-risk indicator and its interactions with year and insurer; county population; the share under age 65 with subsidy-eligible income; and lagged insurer presence. County groups enter additively. The risk dimension is the same predicted spending-risk group used in insurance preferences.

C Details on Expected Claims Cost Prediction

This appendix describes how I construct the spending risk groups used in the demand model and the expected claims costs used in the supply model. Section C.1 describes the former, and Section C.2 describes the latter.

C.1 Predicted spending risk

I construct a spending-risk score to summarize health-related differences in insurance preferences and expected claims costs. The estimation sample pools 421,521 eligible individual-market person-years from 2017–2019. The outcome is $\log(1 + C_{it}/M_{it})$, where C_{it} is total medical and pharmacy spending and M_{it} is covered enrollment months, summed across the individual's in-

surer records within the year.[20]

Predictors and estimation The predictors are an indicator for age 45 or older, sex, and three-digit diagnosis indicators. I restrict diagnosis categories to those available in both the claims data and MEPS so that the scoring rule can be applied consistently across datasets. Categories observed in at least 30 training person-years enter separately; the remaining categories enter through a pooled indicator. The resulting specification includes 419 individual diagnosis indicators and the pooled indicator. I estimate a ridge regression weighted by covered enrollment months, penalizing the diagnosis coefficients while leaving the intercept and demographic coefficients unpenalized. The fitted log-spending score summarizes relative spending risk.

Sample splitting and validation I randomly allocate individuals to training, tuning, and test samples in proportions of 60, 20, and 20 percent, keeping all observations for each individual in the same sample. Diagnosis support is determined using the training sample. I select the ridge penalty from $\{0, 1, 10, 100, 1000\}$ by minimizing weighted prediction error in the tuning sample. With the selected penalty of 100, I refit the model using the combined training and tuning samples and evaluate predictions in the held-out test sample. Across 84,357 test person-years, the weighted R^2 for log spending is 0.598, compared with 0.064 for a model containing only age and sex. The weighted root mean squared error is 1.502.

Risk-group construction After validation, I fix the predictor set and penalty and refit the model using all eligible 2017–2019 observations. Let $\widehat{s}_{it}$ denote the resulting score. I define

$$r_{it} = \mathbf{1}\{\widehat{s}_{it} \geq c\}, \qquad c = 3.3641,$$

where c is the unweighted median score across eligible person-years pooled over 2017–2019. Two groups allow preferences and costs to vary with spending risk while keeping the equilibrium computation tractable. Using a common cutoff preserves the same classification rule across years. Among all scored person-years, the high-risk share is 48.4 percent in 2017, 48.9 percent in 2018,

[20] Concurrent diagnoses allow the score to cover enrollees without observable prior-year claims histories. Dickstein, Ho and Mark (2024) similarly use a concurrent risk measure because consumer churn limits the sample available for a measure based on lagged information.

and 51.9 percent in 2019.

C.2 Type–plan expected claims costs

I estimate type–plan expected claims costs to capture how spending varies across insurers with different networks even conditional on other individual characteristics. The estimation sample contains 419,461 person–insurer–year observations from 252,706 individuals during 2017–2019. The outcome is total medical and pharmacy spending per covered month, weighted by covered enrollment months. To limit the influence of extreme realized expenditures, I cap spending at the member-month-weighted 95th percentile of the training sample; the threshold used for the final fit is $2,002.48 PMPM.

The cost prediction model uses age, sex, spending-risk group, hospital network value, county, rating area, year, insurer, metal tier, and exchange channel. Spending-risk groups follow the pooled-median classification described above. I estimate a flexible regression that accommodates nonlinearities and interactions among these characteristics while producing positive predictions.[21] With the specification fixed, I fit the final model using all eligible observations and evaluate it at each type–plan combination required by the equilibrium model.

I convert predicted total medical spending into the amount the insurer must finance by approximating the insurer's share using the plan's actuarial value. For example, a consumer with $1,000 in expected monthly spending generates $700 in expected insurer payments under a standard silver plan with 70% actuarial value.

More generally, expected insurer claims costs equal the cost of standard coverage plus the unreimbursed cost of additional CSR coverage. Let $\widehat{G}_{kjfm}$ denote predicted gross monthly spending for consumer type k, metal tier j, insurer f, and market m. The standard actuarial value is AV_j^{base}, while AV_{kjm} includes any cost-sharing reductions available to the consumer. With ρ_m denoting the government's reimbursement rate for the additional CSR obligation, expected insurer claims costs are

$$HC_{kjfm} = \widehat{G}_{kjfm} AV_j^{\text{base}} + (1 - \rho_m)\widehat{G}_{kjfm} \left(AV_{kjm} - AV_j^{\text{base}} \right)_+ .$$

I evaluate the cost predictions using five-fold cross-validation, assigning all observations for

[21] I use histogram gradient-boosted regression with Poisson loss, 250 boosting iterations, a learning rate of 0.05, at most 31 leaves per tree, and at least 20 observations per leaf.

each individual to the same fold. Each individual's predicted spending comes from a cost regression estimated on the other four folds, with the spending cap calculated from those training observations. I aggregate observed and predicted spending into cells defined by year, county, rating area, age, sex, exchange channel, insurer, and metal tier, weighting by covered months. As reported in Table 8, for the capped spending outcome, the model explains 44.4% of weighted variation in cell-average spending, with a root mean squared error of $196.21 PMPM and a mean absolute error of $117.43 PMPM. Predicted mean spending is $262.91, compared with $262.87 observed. Absolute mean prediction bias is below 1% within each year, coverage channel, metal tier, age group, and sex.

Table 8: Cross-validation of expected claims cost predictions

Statistic	All validation cells	Cells with at least 10 individuals
Number of cells	65,924	12,300
Median individuals per cell	3	17
Share of matched validation member-months	100.0%	61.8%
Weighted R^2	0.444	0.646
Root mean squared error, PMPM	$196.21	$95.19
Mean absolute error, PMPM	$117.43	$71.46
Actual mean spending, PMPM	$262.87	$232.00
Predicted mean spending, PMPM	$262.91	$234.29
Mean prediction bias	0.02%	0.99%

Notes: Predictions are obtained from five-fold cross-validation at the individual level. Cells are defined by year, county, rating area, age, sex, exchange channel, insurer, and metal tier. Observed spending is capped at the member-month-weighted 95th percentile of the corresponding training sample. Cell averages and validation statistics are weighted by covered months. The matched validation sample contains 403,431 observations and 4,192,225 member-months. The second results column restricts attention to larger cells. Mean prediction bias is the percentage difference between predicted and actual mean spending. PMPM denotes per member per month.

D Details on Marginal Cost Residual Estimation

This appendix describes the two steps used to recover the residual component of marginal cost. I first invert residual marginal cost from observed prices for county-firm-year cells in which the firm is active. Because the medical-loss-ratio (MLR) constraint may be either slack or binding, the inversion constructs a candidate under each regime and retains a cost only when the data select a unique admissible candidate. I then project the retained costs onto observed firm and

market characteristics. The fitted projection supplies residual-cost predictions for firm-market cells without a uniquely inverted cost, while its residuals form the empirical distribution of the post-entry marginal-cost shock.

D.1 Marginal-cost inversion

Let $Q_{fm}(p)$ denote enrollment, $R_{fm}(p)$ premium revenue, and $H_{fm}(p)$ expected claims expenditure for firm f in county-year market m, evaluated at the observed prices. The demand system determines these objects and their derivatives with respect to the firm's own price. I write the residual component of marginal cost as c_{fm}, so variable profit is

$$\pi_{fm}(p, c_{fm}) = R_{fm}(p) - H_{fm}(p) - c_{fm}Q_{fm}(p). \tag{3}$$

The hard MLR constraint requires claims plus residual costs to be at least 80 percent of premium revenue. Defining

$$g_{fm}(p, c_{fm}) = H_{fm}(p) + c_{fm}Q_{fm}(p) - \tau R_{fm}(p), \qquad \tau = 0.80, \tag{4}$$

the constraint is $g_{fm}(p, c_{fm}) \geq 0$.

Because the MLR regime is not observed, I construct two residual-cost candidates for every active county-firm-year cell. Under a slack MLR constraint, the multiplier is zero and the candidate c^S_{fm} solves the ordinary pricing first-order condition,

$$\frac{\partial \pi_{fm}(p, c^S_{fm})}{\partial p_{fm}} = 0, \qquad g_{fm}(p, c^S_{fm}) \geq 0. \tag{5}$$

Under a binding constraint, the candidate is obtained directly from the MLR equality,

$$c^B_{fm} = \frac{\tau R_{fm}(p) - H_{fm}(p)}{Q_{fm}(p)}. \tag{6}$$

I then recover the implied multiplier λ_{fm} from

$$\frac{\partial \pi_{fm}(p, c^B_{fm})}{\partial p_{fm}} + \lambda_{fm}\frac{\partial g_{fm}(p, c^B_{fm})}{\partial p_{fm}} = 0 \tag{7}$$

and require $\lambda_{fm} \geq 0$. Thus, each candidate must satisfy the restrictions associated with the regime under which it was constructed. I also require positive total marginal cost for every available type–plan combination and nonnegative operating profit, $R_{fm} - H_{fm} - c_{fm}Q_{fm} \geq 0$. The profit condition applies to every candidate, including cells with two regime-consistent candidates. It rules out costs under which the observed premium earns a loss before entry costs, despite the option to choose a break-even premium after marginal costs are realized. I retain the cell only when exactly one distinct admissible cost remains and the inversion is sufficiently well determined by the pricing conditions.

The inversion begins with 434 observed active firm–market cells. Constructing two regime-specific values per cell yields 868 candidates, of which 788 satisfy the strict positive-total-cost condition. The procedure retains a unique admissible residual cost in 428 cells. Of the six excluded cells, three retain two admissible candidates, one fails the nonnegative-profit screen, and two have weakly determined inversions. Section D.3 reports sensitivity to reinstating these cells.

D.2 Projection and residual-cost shocks

The second step maps the 428 uniquely inverted costs into residual-cost predictions for the full set of potential firm-market cells and estimates the distribution of the marginal-cost shock. While the inverted $\widehat{c}_{fm}$ all imply positive total marginal cost at the observed prices, they are not constrained to be strictly positive. As a result, if the expected cost component is sufficiently close to zero at some counterfactual prices, the implied total marginal cost could turn negative, which is hard to interpret. For this reason, I instead estimate

$$\log\left(\widehat{c}_{fm} + h_{fm}^{\min}\right) = X_{fm}'\gamma + \omega_{fm}, \tag{8}$$

where $h_{fm}^{\min} = \min_{k,j} HC_{kjfm}$ is the minimum expected claims cost over the firm's supported type–plan combinations in market m. The transformed dependent variable is the logarithm of the smallest total marginal cost that the firm can face among all the possible type-plan cells in a market. Mapping a fitted value or cost shock back according to

$$c_{fm}(\omega) = \exp\left(X_{fm}'\widehat{\gamma} + \omega\right) - h_{fm}^{\min} \tag{9}$$

guarantees $HC_{kjfm} + c_{fm}(\omega) > 0$ for every support-available type and plan because $HC_{kjfm} \geq h_{fm}^{\min}$. The estimated shock is given by

$$\widehat{\omega}_{fm} = \log\left(\widehat{c}_{fm} + h_{fm}^{\min}\right) - X_{fm}'\widehat{\gamma}. \tag{10}$$

The covariates in X_{fm} capture prior market presence and firm-specific cost and demand conditions. They include an indicator for the firm's presence in the same county in the preceding year and the share of other counties in the same rating area in which the firm was present in the preceding year. The remaining continuous covariates summarize the firm's silver-plan primitives in the county: the mean, standard deviation, and 90th percentile of expected claims and the mean price-independent demand utility. These covariates capture the fact that residual marginal costs may vary systematically with the level and dispersion of the population's expected medical needs and with the firm's underlying demand conditions.

Table 9 reports the projection estimates. Because the dependent variable is a residual term, the individual coefficients are hard to interpret. Overall, the regressors and fixed effects jointly explain 82.4 percent of the variation in the transformed residual-cost margin.

Figure 10 plots the resulting distribution of the marginal cost shocks. The distribution is centered close to zero with most observations concentrated in a relatively narrow interval. These residuals form the empirical marginal-cost-shock distribution used by the subsequent cost-draw construction.

To summarize total marginal cost at the same firm–market level, I first average HC_{kjfm} across the firm's support-available plans for each type and then average across types using their population weights. Denoting this average by $\overline{HC}_{fm}$, the cell-level measure is

$$\overline{MC}_{fm} = \overline{HC}_{fm} + \widehat{c}_{fm}. \tag{11}$$

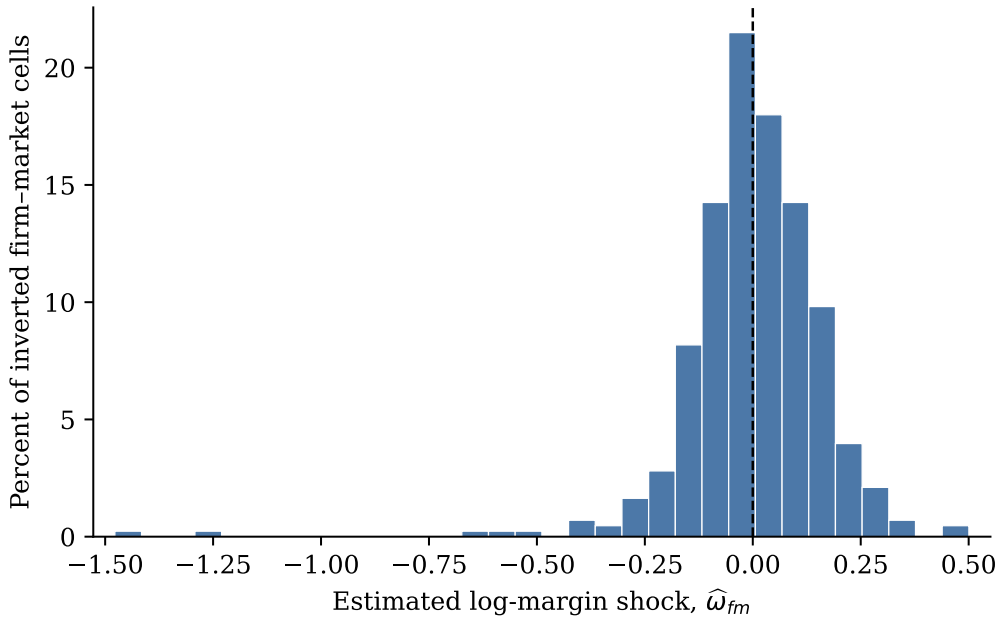
Figure 11 displays the distribution of $\overline{MC}_{fm}$. The median is approximately \$592 per member per month, and the 99th percentile is approximately \$925.

Table 9: Projection of the Residual-Cost Margin

	$\log(\widehat{c}_{fm} + h_{fm}^{\min})$
Lagged presence in the same county	-0.1906^{***}
	(0.0611)
Lagged presence share in other rating-area counties	0.1663^{***}
	(0.0528)
Mean silver-plan expected claims	-0.00027
	(0.00087)
Standard deviation of silver-plan expected claims	-0.00075
	(0.00269)
90th percentile of silver-plan expected claims	0.00030
	(0.00085)
Mean silver-plan price-independent utility	0.0117
	(0.0106)
Firm fixed effects	Yes
County fixed effects	Yes
Year fixed effects	Yes
Observations	428
R^2	0.824
Adjusted R^2	0.786

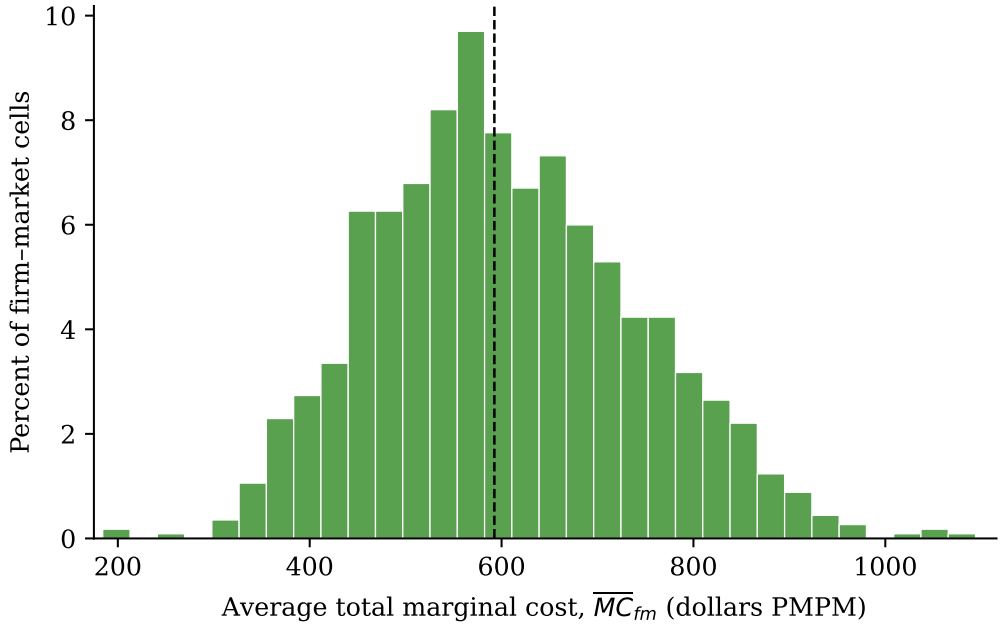
Notes: The table reports unweighted OLS estimates. Conventional OLS standard errors are in parentheses. Expected-claims and utility summaries are computed over the firm's support-available silver plans and consumer types in the county. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Figure 10: Distribution of Estimated Residual-Cost Shocks



Notes: The histogram plots the residuals from equation (8). The dashed vertical line marks zero.

Figure 11: Distribution of Average Total Marginal Cost Across Firm-Market Cells



Notes: For each firm-market cell, predicted claims are averaged across the firm's support-available plans within type and then across types using population weights. The final cell-specific residual-cost estimate is added to this average. The dashed vertical line marks the median.

D.3 Robustness to excluded cells

I assess whether the six excluded county-firm-year cells materially affect the cost projection by reestimating equation (8) with these cells included. Three cells retain both a slack and a binding candidate, two have weakly identified inverted costs, and one has a unique regime-consistent candidate with negative operating profit. I first add the three ambiguous cells using either their slack or their binding candidates. I then repeat both regressions with the other three cells included, using the binding candidates for the two weakly identified cells and the slack candidate for the negative-profit cell. All specifications retain the same covariates and firm, county, and year fixed effects.

Table 10 reports the four alternative regressions. The coefficient estimates and the model fit remain stable across specifications. Relative to the 428-cell baseline in Table 9, the mean absolute changes in projected costs are \$1.93, \$4.47, \$3.64, and \$4.39 per enrollee per month in columns (1)–(4), respectively.

Table 10: Sensitivity of the Cost Projection to Excluded Cells

	(1)	(2)	(3)	(4)
	$\log(\widehat{c}_{fm} + h_{fm}^{\min})$			
Reinstated cells	Three ambiguous		All six	
Ambiguous-cell regime	Slack	Binding	Slack	Binding
Lagged presence in the same county	-0.1873^{***}	-0.2006^{***}	-0.1875^{***}	-0.2008^{***}
	(0.0607)	(0.0609)	(0.0605)	(0.0608)
Lagged presence share in other rating-area counties	0.1670^{***}	0.1782^{***}	0.1685^{***}	0.1798^{***}
	(0.0519)	(0.0521)	(0.0517)	(0.0519)
Mean silver-plan expected claims	-0.00030	-0.00019	-0.00029	-0.00018
	(0.00087)	(0.00087)	(0.00087)	(0.00087)
Standard deviation of silver-plan expected claims	-0.00056	-0.00104	-0.00054	-0.00100
	(0.00268)	(0.00269)	(0.00267)	(0.00268)
90th percentile of silver-plan expected claims	0.00026	0.00035	0.00025	0.00034
	(0.00085)	(0.00085)	(0.00085)	(0.00085)
Mean silver-plan price-independent utility	0.0110	0.0111	0.0116	0.0117
	(0.0105)	(0.0106)	(0.0104)	(0.0105)
Firm fixed effects	Yes	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Observations	431	431	434	434
R^2	0.823	0.823	0.824	0.824
Adjusted R^2	0.786	0.786	0.788	0.787

Notes: All columns report unweighted OLS estimates with conventional OLS standard errors in parentheses. Columns (1)–(2) add the three ambiguous cells to the baseline sample. Columns (3)–(4) also include the two weakly identified cells using their binding candidates and the negative-profit cell using its slack candidate. Covariates are defined as in Table 9. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

E Entry Estimation Details

I estimate the fixed-cost distribution in two stages following the two-stage logic of Pakes, Ostrovsky and Berry (2007). The first stage estimates entry probabilities and uses them to average pricing-stage profits over possible rival configurations. The second stage estimates fixed costs by matching the probability that expected variable profit covers the fixed-cost draw with observed participation decisions. This appendix details the implementation used in each stage.

E.1 First stage: entry policy function estimation

The estimation sample contains six potential insurers over 2017–2019. An insurer participates if it has at least five enrollees. Prior participation uses the same threshold in the preceding year. Market size is the number of potential individual-market consumers.

Selection and imperfect competition make a firm's entry incentives depend on its own characteristics and those of potential rivals in ways that can be high-dimensional and nonlinear. I start by constructing a candidate library containing a large number of variables describing consumer appeal, expected claims costs, market size, network-related differences, and prior participation. Firm-level summaries include distributions across consumer types and comparisons with the other five potential insurers.

To select a parsimonious subset of state variables, I use a neural network to screen characteristics for their ability to predict the precomputed pricing-stage profits. The five rival-entry indicators enter the neural prediction model as context controls, allowing a characteristic's relevance for profits to depend on the competitive environment, but do not enter the downstream model.[22]

After the neural screening step selects the 40 highest-ranked characteristics that predict profit, I use a penalized Logit to select a smaller subset of these characteristics that best predict observed entry. At each step, I add the variable producing the greatest reduction in four-fold, county-grouped cross-validation log loss. Selection stops when the next variable improves log loss by less than 0.001.

With the seven selected characteristics collected in x_i and $i = (f, m, t)$, I estimate the following

[22] The neural model uses sparsity gates and eight repeated 75/25 training–validation splits. All firms and rival configurations within a county–year remain in the same split. The ranking combines gate importance, selection stability, and contribution to held-out profit prediction.

Logit on the full entry sample:

$$\widehat{q}_i = \Lambda\big(\widehat{a}_0 + \widehat{a}_M \log M_i + \widehat{a}_L L_i + \widehat{a}_{2018}\mathbf{1}\{t = 2018\} + \widehat{a}_{2019}\mathbf{1}\{t = 2019\} + x_i'\widehat{a}_x\big)\,,$$

where Λ is the logistic cumulative distribution function and $L_i = \mathbf{1}\{N_{fm,t-1} \geq 5\}$. The final fit is unpenalized and contains 11 predictors plus an intercept. These probabilities estimate the participation policies played in the observed equilibrium and supply the weights for rival configurations in the next stage.

The seven selected characteristics summarize the nonprice appeal and expected claims costs of potential rivals. Their construction proceeds in two steps. The first step calculates a characteristic for each of the six potential insurers in a county–year. The second step then summarizes that characteristic across the five potential rivals of the focal insurer.

I use nonprice utility and its distribution across consumer types to summarize each rival's appeal and how that appeal differs across consumers. I use population-weighted means or percentiles of mean or maximum insurer plan utility across consumer types to summarize the distribution of an insurer's appeal across potential consumer types. Measures of how insurers compare in consumer appeal either subtract the simple average of the corresponding statistic across all six potential insurers or express the statistic as its percentile rank among those insurers.

To allow the predictive relevance of rival characteristics to depend on the degree of substitution between insurers, I interact them with a measure of overlap between pairs of insurers and include both averages and maxima of these interactions as candidate summaries of the competitive environment. To calculate overlap, let $s_{k,fj}$ denote the probability that consumer type k chooses insurer f between the plans of insurers f and j when premiums are zero for all their plans. The overlap measure is defined as

$$o_{fj} = \sum_k w_k\, s_{k,fj}(1 - s_{k,fj}),$$

where w_k is the type's share of the potential consumer population. Each type contributes most when its choice probability is one-half and contributes little when that probability is close to zero or one.

For a rival characteristic x_j, the overlap-weighted mean is

$$\frac{\sum_{j \neq f} o_{fj} x_j}{\sum_{j \neq f} o_{fj}}.$$

This summarizes the characteristic across the five potential rivals while placing greater weight on those with similar nonprice appeal to the focal insurer. I also construct the largest overlap–characteristic product,

$$\max_{j \neq f} \{o_{fj} x_j\},$$

which selects the largest product of a rival's characteristic and its overlap with the focal insurer.

Table 11 describes the construction of each selected predictor.

Table 11: Selected characteristics in the first-stage entry policy

Selected predictor	Meaning	Construction
Upper-tail plan appeal	Rivals' relative upper-tail appeal, emphasizing those with similar consumer appeal.	Average utility across each insurer's plans for each type. Take the 99th percentile across types, rank among six insurers, then calculate the overlap-weighted mean across rivals.
Peak silver-plan appeal and overlap	Relative peak silver-plan appeal, interacted with similarity in consumer appeal.	Find each insurer's maximum silver utility across types and rank among six insurers. Multiply each rival's rank by overlap and take the largest product.
Lowest average best-plan appeal	The lowest average appeal of a rival's best available plans, relative to potential insurers.	Take each insurer's maximum plan utility for each type. Average across types, subtract the six-insurer mean, then take the lowest rival value.
Silver-utility dispersion and overlap	Variation in rivals' silver-plan appeal across consumers, interacted with overlap.	Calculate each rival's 95th–5th percentile spread in silver utility across types. Multiply by overlap and take the largest product.
Lower-tail silver claims and overlap	Relative claims costs toward the lower end of rivals' cost distributions, interacted with overlap.	Calculate each insurer's 25th percentile of silver claims across types and subtract the six-insurer mean. Multiply each rival's value by overlap and take the largest product.
Median rival's upper-tail silver claims	A typical rival's relative upper-tail claims costs.	Calculate each insurer's 95th percentile of silver claims across types and subtract the six-insurer mean. Take the median rival value.
Second-lowest peak silver-plan appeal	The lower end of rivals' relative peak silver-plan appeal.	Find each insurer's maximum silver utility across types and subtract the six-insurer mean. Take the second-lowest rival value.

Notes: Measures are constructed within county–year. Utility excludes premiums; claims are expected claims. Means and percentiles across types use population weights. Ranks and six-insurer means compare the same statistic across all potential insurers, weighted equally. Rival summaries use the other five insurers regardless of participation. With overlap o_{fj} defined above, the overlap-weighted mean is $\sum_{j \neq f} o_{fj} x_j / \sum_{j \neq f} o_{fj}$, and the largest product is $\max_{j \neq f} \{o_{fj} x_j\}$.

E.2 First-stage fit and validation

The first-stage probabilities determine how frequently a potential entrant expects to face each rival. I assess their fit by comparing observed and predicted participation across market-size thirds and prior-participation groups. Table 12 shows that the fitted probabilities capture the large participation gap between previously absent and previously present insurers and remain close to observed rates within market-size groups.

Table 12: First-stage entry fit by market size and prior participation

Market size	Prior participation	Observations	Observed (%)	Fitted (%)
Small	Absent	279	9.3	7.5
Small	Present	87	73.6	78.9
Middle	Absent	263	6.8	7.7
Middle	Present	103	88.3	84.9
Large	Absent	171	14.0	15.8
Large	Present	195	88.7	87.9

Notes: Fitted rates average the full-sample first-stage Logit probabilities within each group.

I next assess whether using first-stage probabilities estimated on other counties materially changes the fixed-cost estimates. I divide counties into four folds, estimate the first-stage Logit on three folds, and predict participation in the remaining fold, keeping all years and insurers within a county together. I repeat this procedure using three random partitions, holding the selected predictors fixed. For each partition, I recompute expected variable profits using the held-out probabilities and re-estimate fixed costs. Table 13 shows that the estimates remain stable: the market-size elasticity and prior-participation cost reduction remain close to their baseline values, while mean absolute changes in estimated fixed costs range from 0.9% to 4.4%.

Table 13: Stability of fixed-cost estimates using held-out first-stage probabilities

Statistic	Full-sample probabilities	Held-out probabilities
Market-size elasticity	0.436	0.435–0.446
Prior-participation cost reduction (%)	79.7	80.0–81.0
Mean absolute fixed-cost change (%)	0.0	0.9–4.4

Notes: Held-out results report ranges across three four-fold county partitions. Predictor standardization and Logit coefficients are re-estimated within each training sample, holding the full-sample predictor selection fixed. All fixed-cost estimates use the same second-stage sample, pricing-stage payoff schedule, and half-normal maximum-likelihood specification. The final row averages the absolute percentage change in estimated mean fixed costs relative to the baseline across insurer–county–year observations.

E.3 Second stage: estimation of the fixed cost parameters

I calculate expected variable profit by first averaging over marginal-cost shocks within each rival configuration and then averaging across configurations using the first-stage probabilities. Let Y_i denote observed entry. For each of the $2^5 = 32$ rival configurations $a_{-f} \in \{0,1\}^5$, the cost-shock-averaged profit conditional on own entry is

$$\overline{V}_i(a_{-f}) \equiv \mathbb{E}_\omega \left[V_{fmt}(1, a_{-f}, \omega_{mt}) \right],$$

where the expectation is approximated by the equal-weight mean of accepted pricing solutions for that configuration. Appendix F describes the pricing solver. If $\widehat{q}_{jmt}$ is rival j's fitted first-stage entry probability, the policy-weighted expected variable profit used in the second stage is

$$\widehat{V}_i = \sum_{a_{-f} \in \{0,1\}^5} \overline{V}_i(a_{-f}) \prod_{j \neq f} \widehat{q}_{jmt}^{a_j} (1 - \widehat{q}_{jmt})^{1-a_j}. \tag{12}$$

Fixed costs follow the strict half-normal specification in the main text, with zero location and year effects common to previously present and absent firms. Let M_i denote potential market size and $L_i = \mathbf{1}\{N_{fm,t-1} \geq 5\}$ indicate prior participation. The fixed-cost distribution and its scale are

$$\begin{aligned} FC_i &= \kappa_i |\epsilon_i|, \qquad \epsilon_i \sim N(0,1), \\ \log \kappa_i &= \mu + \rho \log M_i + \delta L_i + \lambda_{t(i)}, \qquad \lambda_{2017} = 0. \end{aligned}$$

The model-implied entry probability is the probability that fixed cost is below expected variable profit. For $\theta = (\mu, \rho, \delta, \lambda_{2018}, \lambda_{2019})$, this probability is

$$p_i(\theta) = \begin{cases} 2\Phi\left(\widehat{V}_i / \kappa_i(\theta)\right) - 1, & \widehat{V}_i > 0, \\ 0, & \widehat{V}_i \leq 0, \end{cases} \tag{13}$$

where Φ is the standard normal cumulative distribution function. Mean fixed cost is $\kappa_i \sqrt{2/\pi}$, so ρ is its market-size elasticity and $\exp(\delta)$ is the ratio of mean fixed costs for previously present insurers to those for previously absent insurers, holding market size and year fixed.

MLE (baseline). I estimate the fixed-cost parameters by maximizing the Bernoulli log likelihood of observed participation,

$$\ell_N(\theta) = \sum_{i=1}^{N} \{Y_i \log p_i(\theta) + (1 - Y_i) \log[1 - p_i(\theta)]\}.$$

The expected profits $\widehat{V}_i$ are held fixed at their first-stage estimates during this maximization.

Inference. The reported 95% confidence intervals are percentile intervals from 1,000 bootstrap replications that resample county panels, refit the first-stage scaler and Logit, recompute expected profits, and re-estimate the second-stage parameters on the same bootstrap samples. All years and insurers within a county are resampled together, and the continuous GMM instruments are restandardized within each replication. The confidence intervals are conditional on the precomputed pricing-stage payoff schedule and the selected first-stage features.

E.4 Robustness to the second-stage estimator

As a robustness check, I estimate the same half-normal fixed-cost specification using GMM, holding expected variable profits and the sample fixed. I use moments that match average entry and its relationship with market size, prior participation, year, and profitability. To capture the competitive environment, I include two summaries of the pricing-stage payoff schedule that do not depend on the estimated first-stage probabilities. The first is average profit across rival configurations,

$$P_i = \frac{1}{32} \sum_{a_{-f} \in \{0,1\}^5} \overline{V}_i(a_{-f}).$$

Second, define the average loss in the focal firm's profit from adding one rival,

$$C_i = \frac{1}{5 \times 16} \sum_{j \neq f} \sum_{a_{-(f,j)} \in \{0,1\}^4} \left[\overline{V}_i(a_j = 0, a_{-(f,j)}) - \overline{V}_i(a_j = 1, a_{-(f,j)}) \right]. \tag{14}$$

Thus, P_i averages all 32 rival states with weight $1/32$, while C_i averages the 80 absent-versus-present comparisons generated by five rivals and 16 configurations of the remaining four rivals. The nine instruments are a constant, log expected variable profit $\log \widehat{V}_i$, the firm's lagged enrollment share, prior participation L_i, log market size $\log M_i$, indicators for 2018 and 2019, and the

two payoff summaries P_i and C_i. Lagged enrollment share is the firm's prior-year enrollment divided by total prior-year enrollment across the six potential insurers in the county. Continuous instruments are centered and scaled to unit standard deviation. Interacting these instruments with the entry residual $Y_i - p_i(\theta)$ gives nine moments for five parameters and four overidentifying restrictions. The reported GMM estimator minimizes the one-step, identity-weighted criterion.

Table 14 shows that both estimators imply fixed costs that rise less than proportionally with market size and are substantially lower for previously present insurers. These qualitative conclusions are therefore robust to the choice of second-stage estimator.

Table 14: Fixed-Cost Estimates: Maximum Likelihood and GMM

	MLE (baseline)		GMM	
Parameter	Estimate	95% CI	Estimate	95% CI
Market-size elasticity ρ	0.436	[0.267, 0.611]	0.283	[0.156, 0.457]
Prior enrollment ≥ 5: δ	-1.595	[-2.029, -1.189]	-1.657	[-2.366, -1.145]
Log-scale constant μ	7.991	[6.973, 9.085]	8.094	[6.993, 9.180]

Notes: The fixed-cost sample contains observations for six potential insurers over 2017–2019. Entry is defined as at least five enrollees, and prior presence uses the same threshold for prior-year enrollment. Year fixed effects are included but not reported. Confidence intervals are the 2.5th and 97.5th percentiles from 1,000 bootstrap replications. The intervals condition on the selected first-stage predictors and precomputed pricing-stage payoffs.

F Equilibrium Solve Details

This appendix describes the algorithm used to solve the pricing equilibrium for each county-year market, entrant set, and marginal-cost draw. The procedure has four main steps. First, it constructs an initial price vector from a previously solved, economically similar case. Second, it searches for a solution to the firms' exact first-order and hard medical-loss-ratio (MLR) conditions. Third, if the exact search fails, it uses coordinate-wise best responses to find an economically stable price vector. Finally, it removes firms with fewer than one expected enrollee and repeats the pricing calculation for the remaining firms. A result is retained only if it satisfies either the exact constrained first-order conditions or the best-response acceptance criteria described below.

F.1 Pricing problem recap

Let $Q_{fm}(p)$, $R_{fm}(p)$, and $H_{fm}(p)$ denote enrollment, gross premium revenue, and expected claims for firm f in county-year market m. Let c_{fm} denote its residual marginal cost. Profit is

$$\pi_{fm}(p) = R_{fm}(p) - H_{fm}(p) - c_{fm}Q_{fm}(p). \tag{15}$$

The MLR constraint is

$$g_{fm}(p) = H_{fm}(p) + c_{fm}Q_{fm}(p) - 0.80R_{fm}(p) \geq 0. \tag{16}$$

F.2 Constrained first-order conditions

For each active firm, the solver considers whether the MLR constraint is slack or binding. If it is slack, the firm's own-price profit derivative must equal zero and the constraint must remain feasible:

$$\frac{\partial \pi_{fm}(p)}{\partial p_{fm}} = 0, \qquad g_{fm}(p) \geq 0. \tag{17}$$

If the constraint binds, the solver requires

$$g_{fm}(p) = 0, \qquad \lambda_{fm} = -\frac{\partial \pi_{fm}(p)/\partial p_{fm}}{\partial g_{fm}(p)/\partial p_{fm}} \geq 0. \tag{18}$$

F.3 Initial values and solution order

I first solve each entrant set with no marginal cost shock as a warm start value for the later cases with different marginal cost shocks. Then, for each entry set, the marginal-cost draws are ordered so that successive cost vectors are close. Each draw begins from a previously solved draw for the same entry set.

F.4 Constrained first-order solution search

The solver first checks whether the initial prices already satisfy the constrained first-order conditions. It then tests the binding set inherited from the warm-start source. If that binding set yields an exact solution, the search ends. Otherwise, the solver constructs a numerical warm start by

solving smoothed Fischer–Burmeister complementarity equations (Fischer, 1992).[23]

If none of the warm-start binding sets yield a solution, the solver then enumerates the remaining possible sets of binding MLR constraints among the active firms, ordered by (1) increasing number of binding firms; and (2) firm ID within sets of the same size. For every candidate binding set, the solver first tries to find prices that solve the corresponding system of binding MLR equations and slack first-order conditions. After optimization, the solver verifies the slackness classification by checking MLR feasibility, slack-firm profit derivatives, binding-firm MLR gaps, implied MLR multipliers, and constraint multiplier nonnegativity. The candidate is accepted only if the maximum violation (normalized by enrollment) across all these conditions is at most 0.001. The solver stops at the first valid binding set.

F.5 Best-response resolution

If no binding set produces an acceptable hard-KKT solution, the solver searches for profitable unilateral price deviations. A deviation is economically material when its profit gain exceeds

$$\min\{\$100,\ 0.01\max(0, \text{current profit})\}. \tag{19}$$

When a material profitable deviation exists, the solver updates the firm with the largest feasible gain and repeats. A result is classified as solved with a best-response resolution if every firm is best responding to the other firms with a feasible price and no firm has a material profitable deviation.

[23] For each active firm, the least-squares solver jointly chooses prices and MLR multipliers to minimize the residuals from the stationarity equation

$$\frac{\partial \pi_{fm}(p)}{\partial p_{fm}} + \lambda_{fm}\frac{\partial g_{fm}(p)}{\partial p_{fm}} = 0$$

and the Fischer–Burmeister equation

$$\phi\Big(\lambda_{fm}, g_{fm}(p)\Big) = \sqrt{\lambda_{fm}^2 + g_{fm}(p)^2} - \lambda_{fm} - g_{fm}(p) = 0.$$

The Fischer–Burmeister equation holds if and only if $\lambda_{fm} \geq 0$, $g_{fm}(p) \geq 0$, and $\lambda_{fm} g_{fm}(p) = 0$, thereby imposing MLR feasibility and complementary slackness without specifying in advance whether each firm's constraint binds. Numerically, the solver replaces ϕ with the smooth approximation

$$\phi_\tau\Big(\lambda_{fm}, g_{fm}(p)\Big) = \sqrt{\lambda_{fm}^2 + g_{fm}(p)^2 + \tau^2} - \lambda_{fm} - g_{fm}(p)$$

and progressively reduces τ to approach the discrete solution.

F.6 Pruning

After a pricing solution is obtained, all active county-firm cells with expected enrollment strictly below one are removed simultaneously. The reduced entry set is then solved again, starting from the preceding prices, until all the remaining firms reach an acceptable solution.